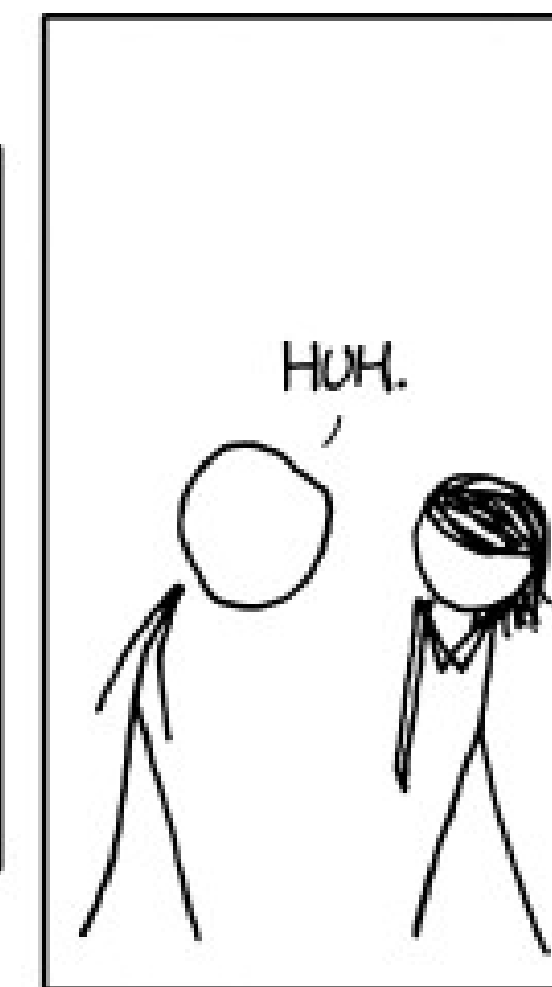
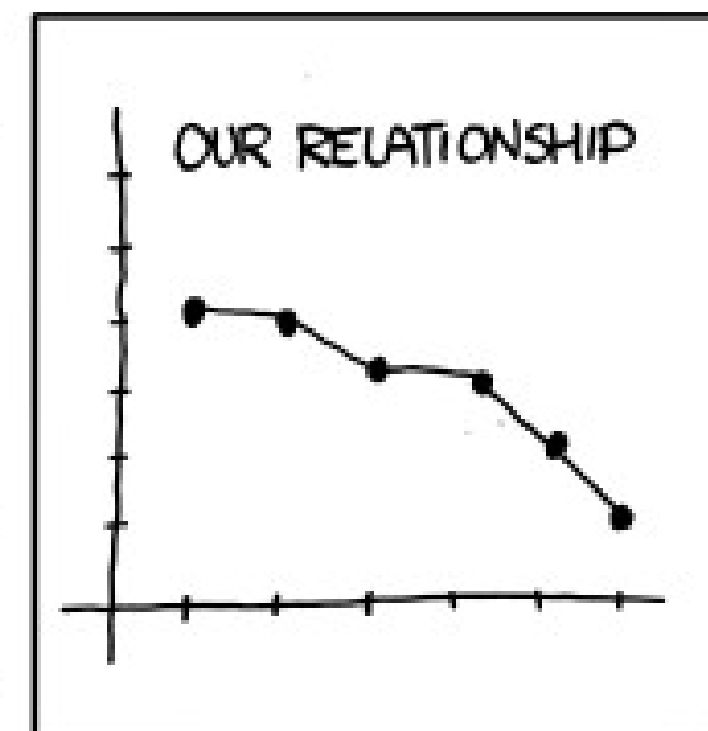
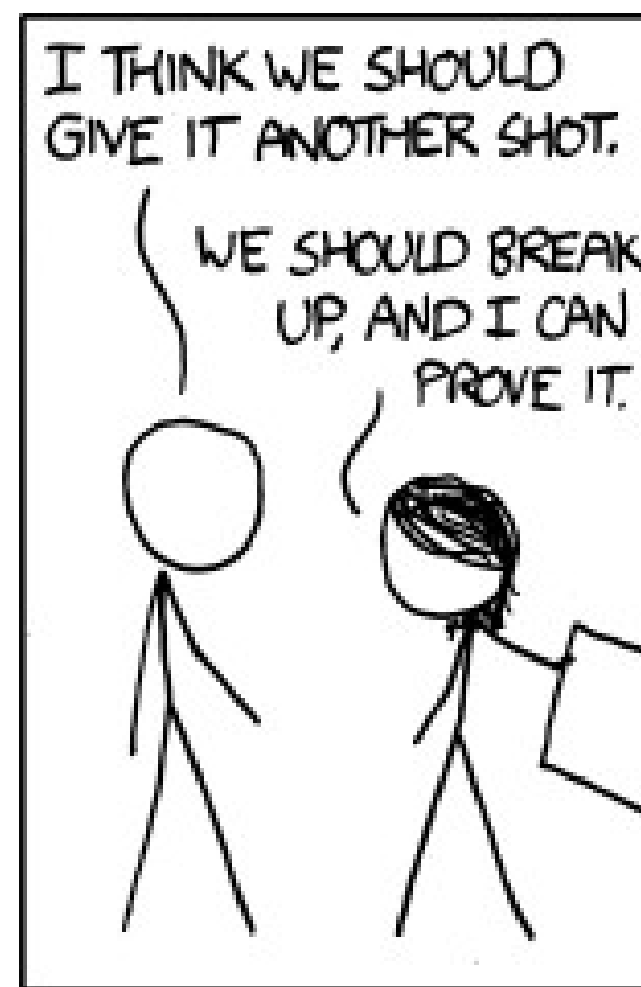


Perception: Part I



Sarah Groves

based on materials by Hannah Bako, Nur Yildirim, Adam Perer, and Zhicheng Liu

How this fits in

- You've learned about **effectiveness** and **expressiveness**
- **Effectiveness** is based on accuracy but also other criteria dependent on how our eyes **perceive** and our brains **interpret** information
- How are the channel rankings justified?
 - Speed and accuracy of interpreting information 
 - Discriminability
 - Separability
 - Popout
 - Relative vs absolute judgments
- To understand these justifications, we must understand ***how the eye perceives visual information***

Perception vs. Cognition

Perception

- Identification and interpretation of sensory information
- From the physical stimulus to recognizing information
- Shaped by learning, memory, expectation

Perception vs. Cognition

Perception

- Identification and interpretation of sensory information
- From the physical stimulus to recognizing information
- Shaped by learning, memory, expectation

Cognition

- The processing of information, applying knowledge

Perception:

- Hear someone speak

Cognition

- Understand the language and the words

What do you hear?
What do you hear after being
primed with a word?

<https://www.facebook.com/watch/?v=734729034059779>

What do you hear?
What do you hear once you know
the sentence?

<https://philiprubin.me/sws-sentences>

Perception is what we hear immediately; a reflex
Cognition is what we understand to hear; this takes more thought

Perception vs. Cognition (Visual System)

Perception

Eye, optical nerve, visual cortex

Basic perception

First processing
(edges, planes)

Not conscious

Reflexes

Cognition

Recognizing objects

Relations between objects

Conclusion drawing

Problem solving

Learning

...

Why is perception important for data viz?

- Choosing effective visual encodings requires knowledge of how human visual perception works
- Evaluation (e.g. user studies) only tells us which design works better, not WHY.
- Theories or models on human perception explain why certain designs succeed or fail.

How many 3's?

**1281768756138976546984506985604982826762
9809858458224509856458945098450980943585
9091030209905959595772564675050678904567
8845789809821677654876364908560912949686**

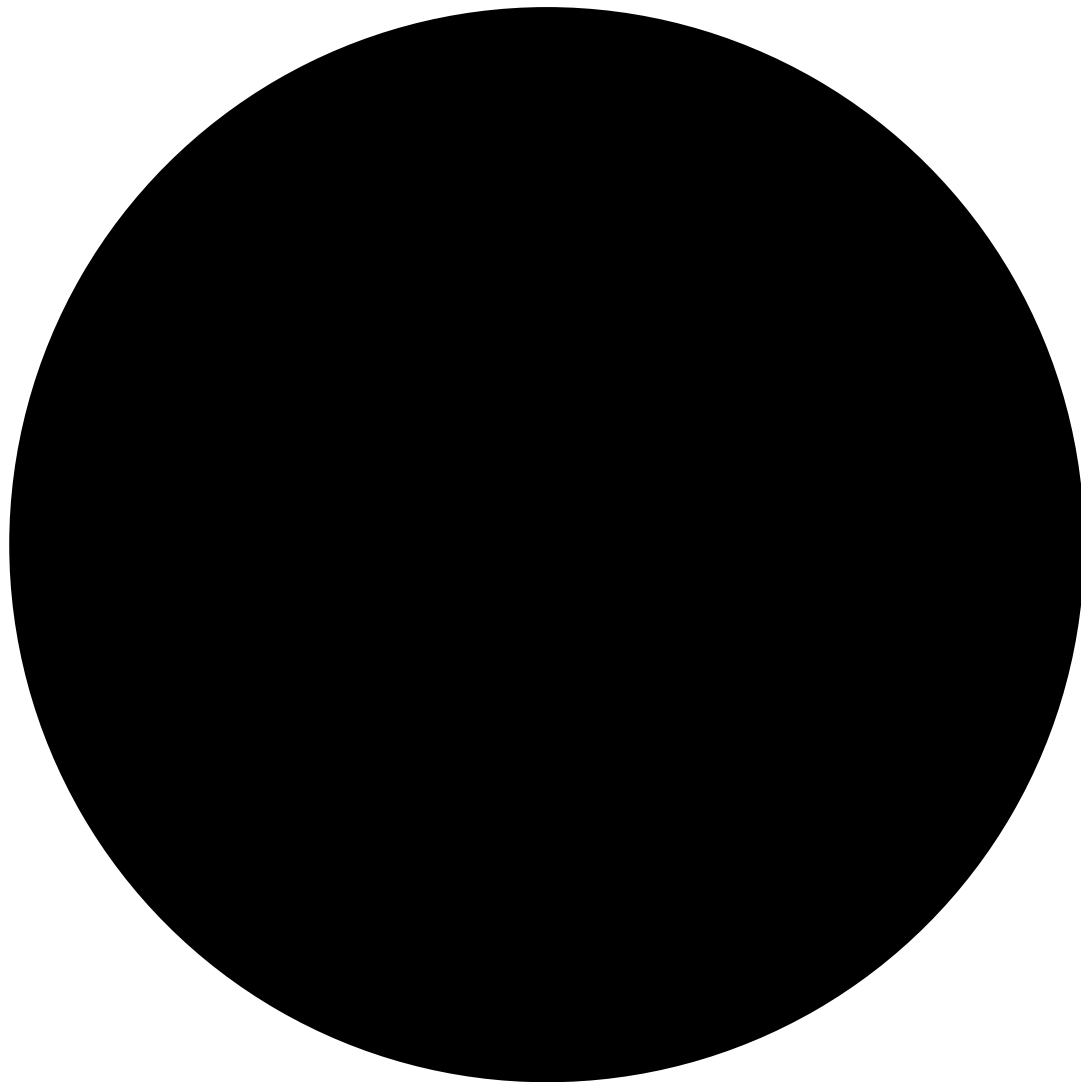
10 seconds

Slide courtesy of John Stasko

How many 3's?

12817687561**3**8976546984506985604982826762
980985845822450985645894509845098094**3**585
90910**3**0209905959595772564675050678904567
8845789809821677654876**3**64908560912949686

10 seconds



Slide courtesy of John Stasko

How many 3's?

12817687561**3**8976546984506985604982826762
980985845822450985645894509845098094**3**585
90910**3**0209905959595772564675050678904567
8845789809821677654876**3**64908560912949686
1298457689567489567458978869579456745794
0856497146546245968945604567765406745790
1221987549649590129850968907128790989569

10 seconds

Take Home Point

Vision is “constructed” top down from the input

“What you see when you see a thing depends on what the thing is. What you see the thing as depends on what you know about what you are seeing.”

**Zenon Pylyshyn
Cognitive Scientist and
Philosopher**

**We want to make the most
important things easy to
perceive automatically.**

Via “pre-attentive processing”

What are the most important things? → Goal of the visualization
How do we make sure they are perceived quickly? → perceptive criteria

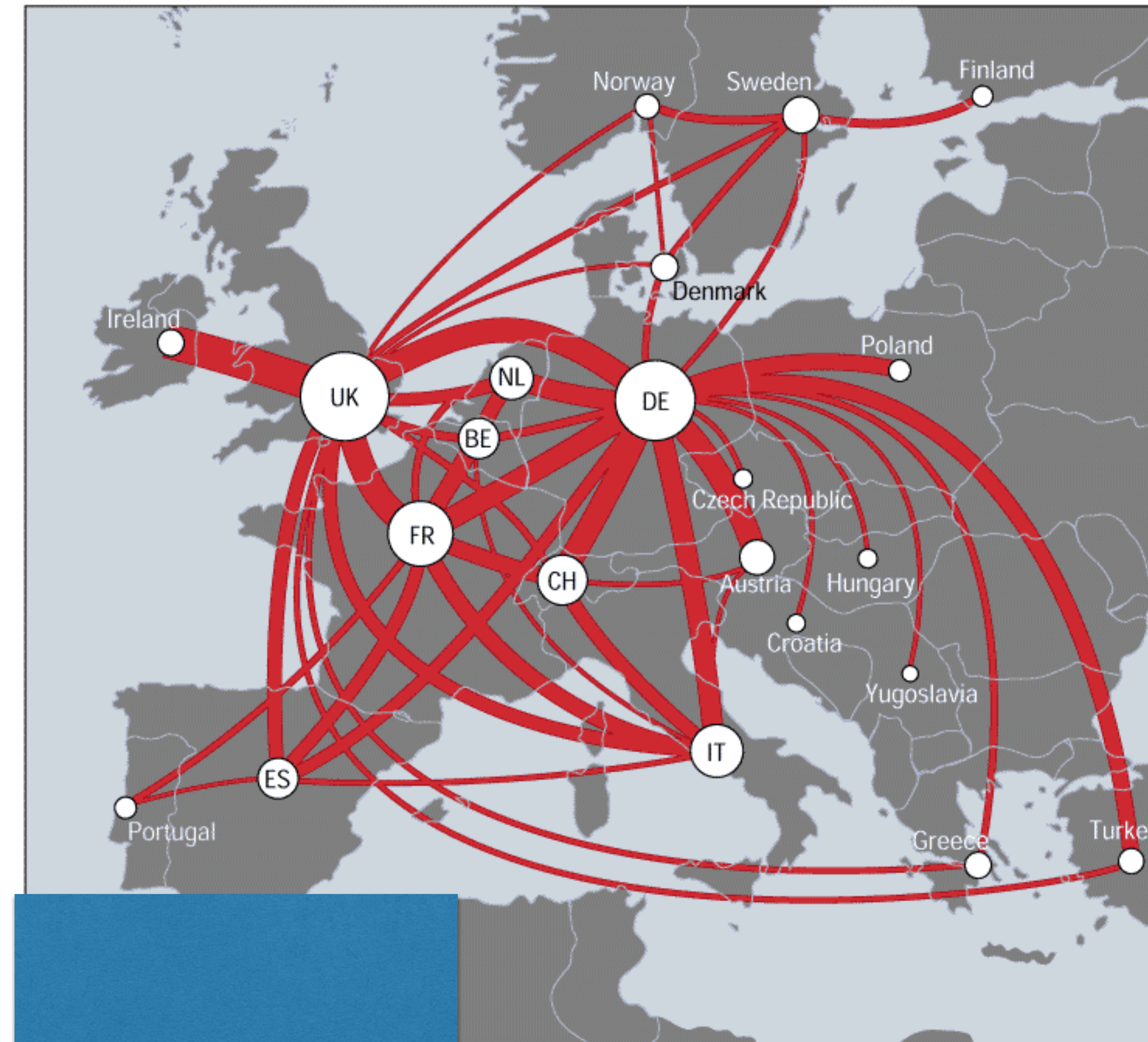
Channel effectiveness

- accuracy: how precisely can we tell the difference between encoded items?
- discriminability: how many unique steps can we perceive?
- separability: is our ability to use this channel affected by another one?
- popout: can things jump out using this channel?

Discriminability

Discriminability: How many usable steps?

- must be sufficient for number of attribute levels to show
 - linewidth: few bins

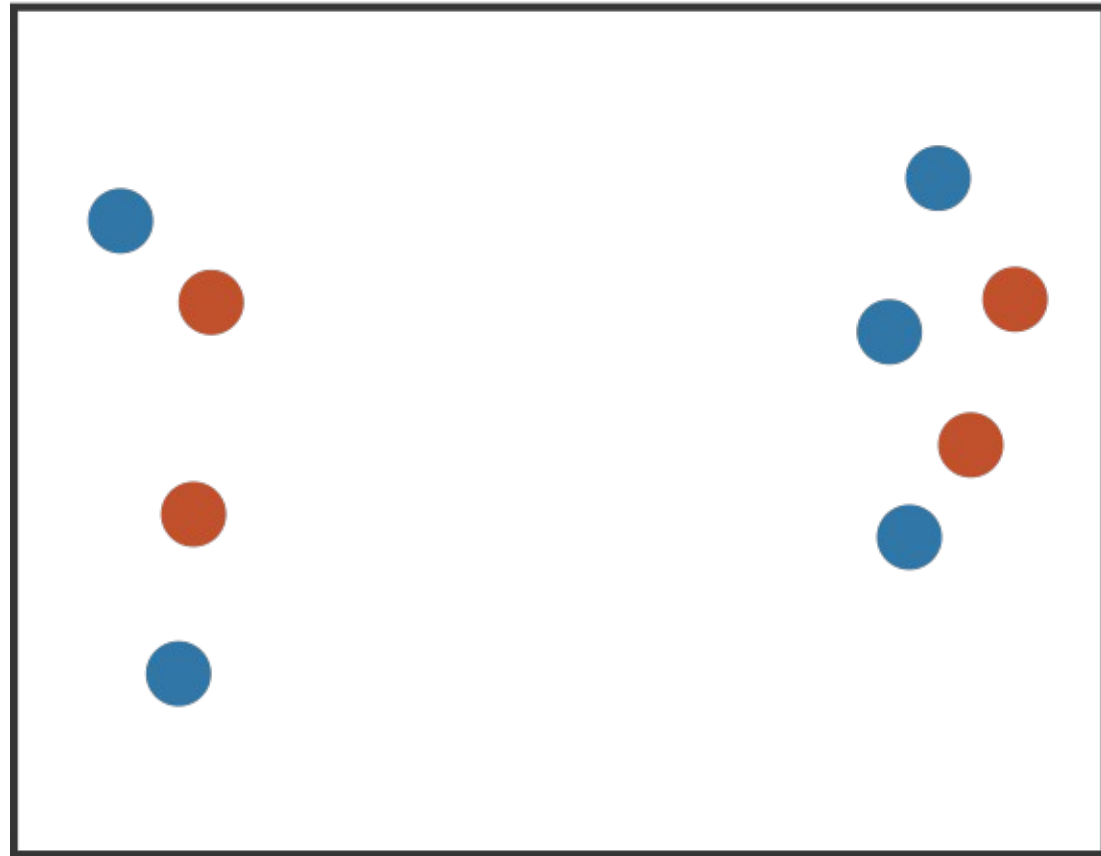


Separability

Separability vs. Integrality

Position

+ Hue (Color)

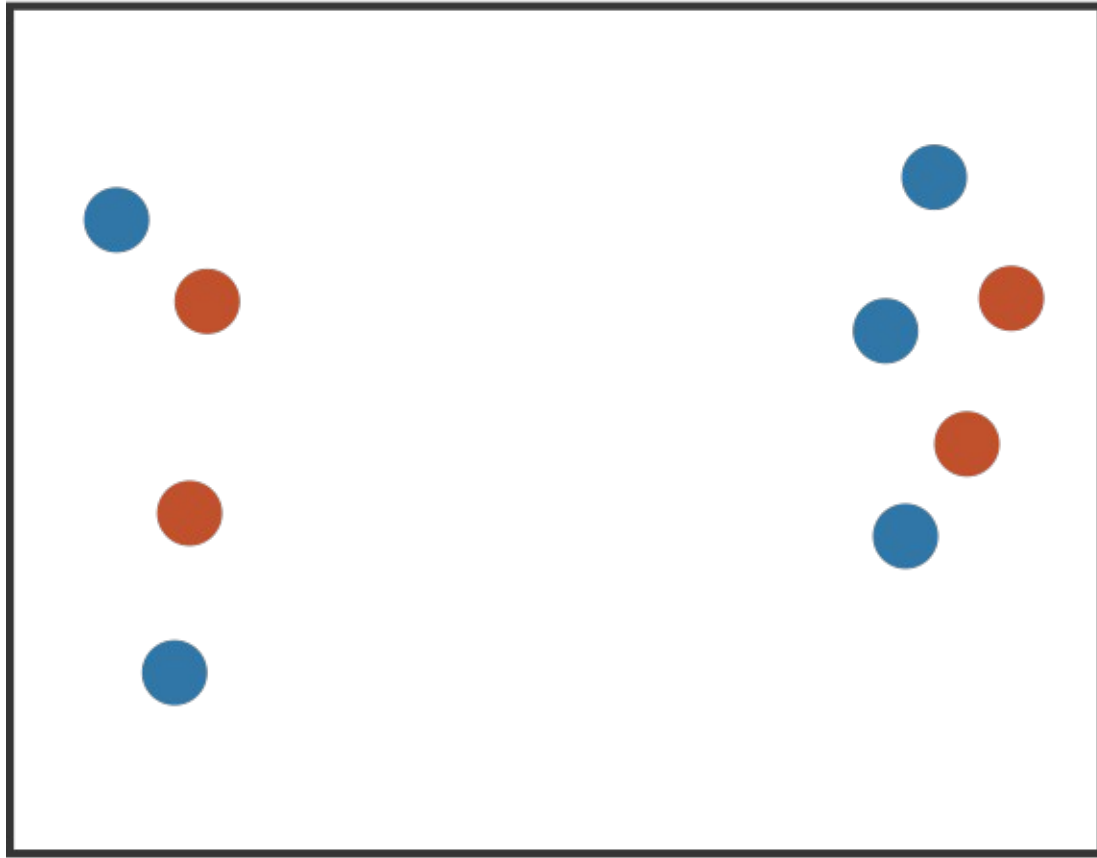


Fully separable

2 groups
each

Separability vs. Integrality

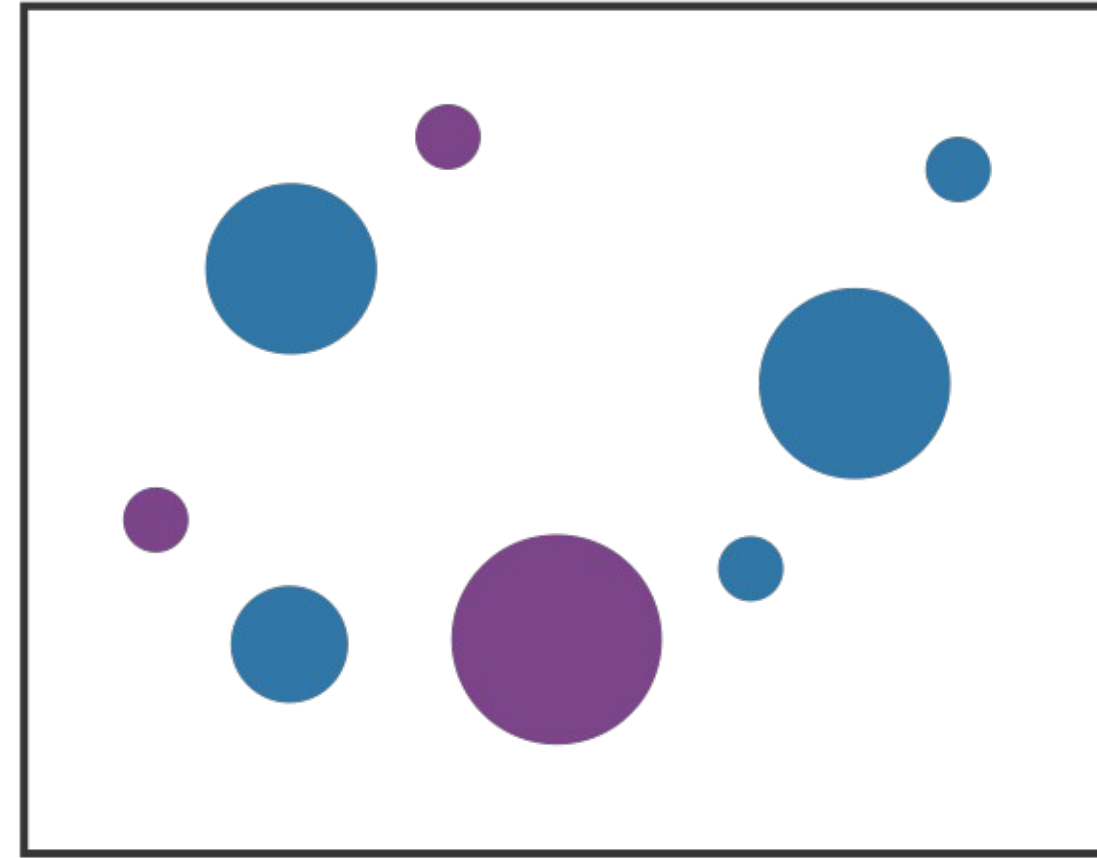
Position
+ Hue (Color)



Fully separable

2 groups
each

Size
+ Hue (Color)

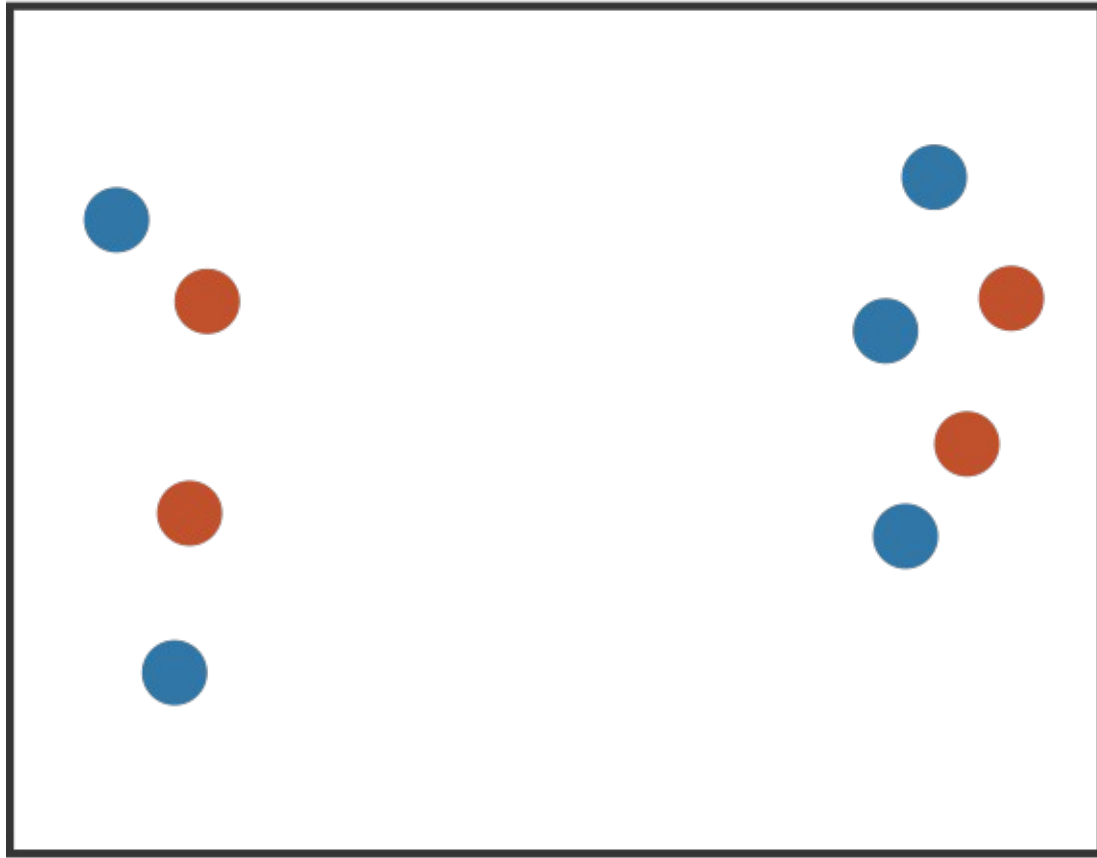


Some interference

2 groups
each

Separability vs. Integrality

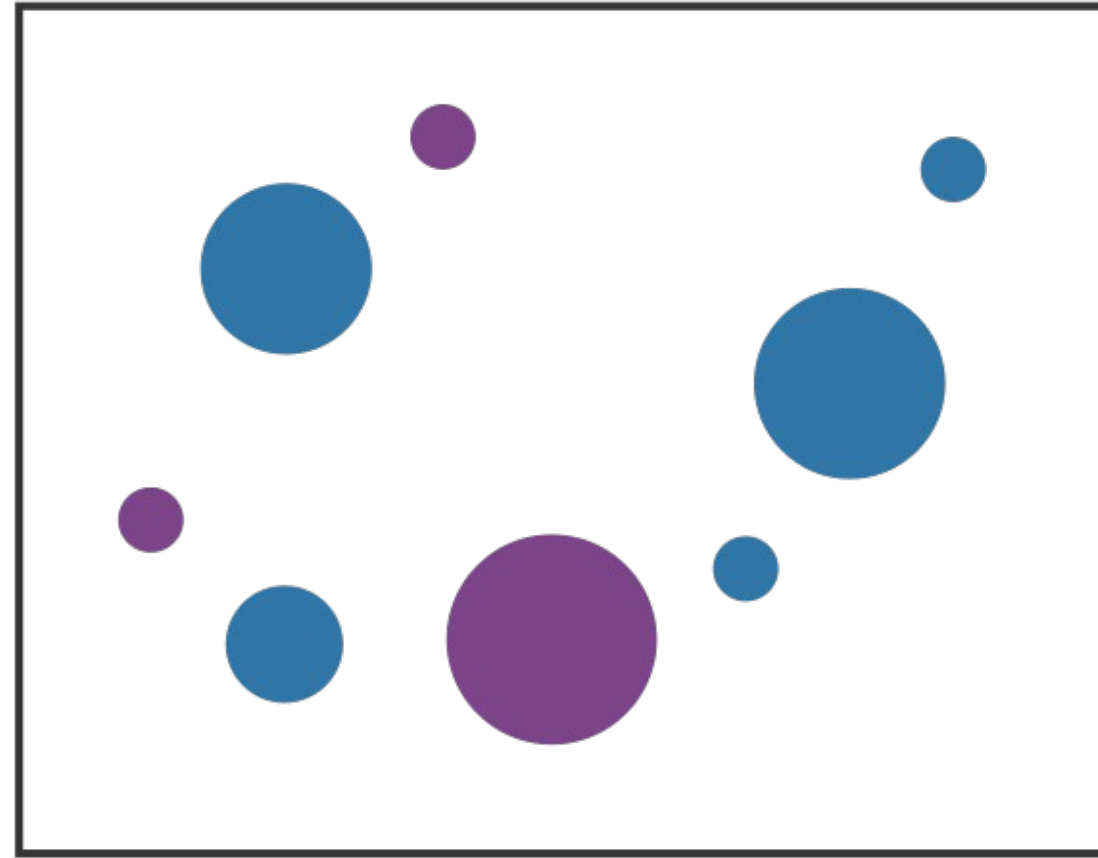
Position
+ Hue (Color)



Fully separable

2 groups
each

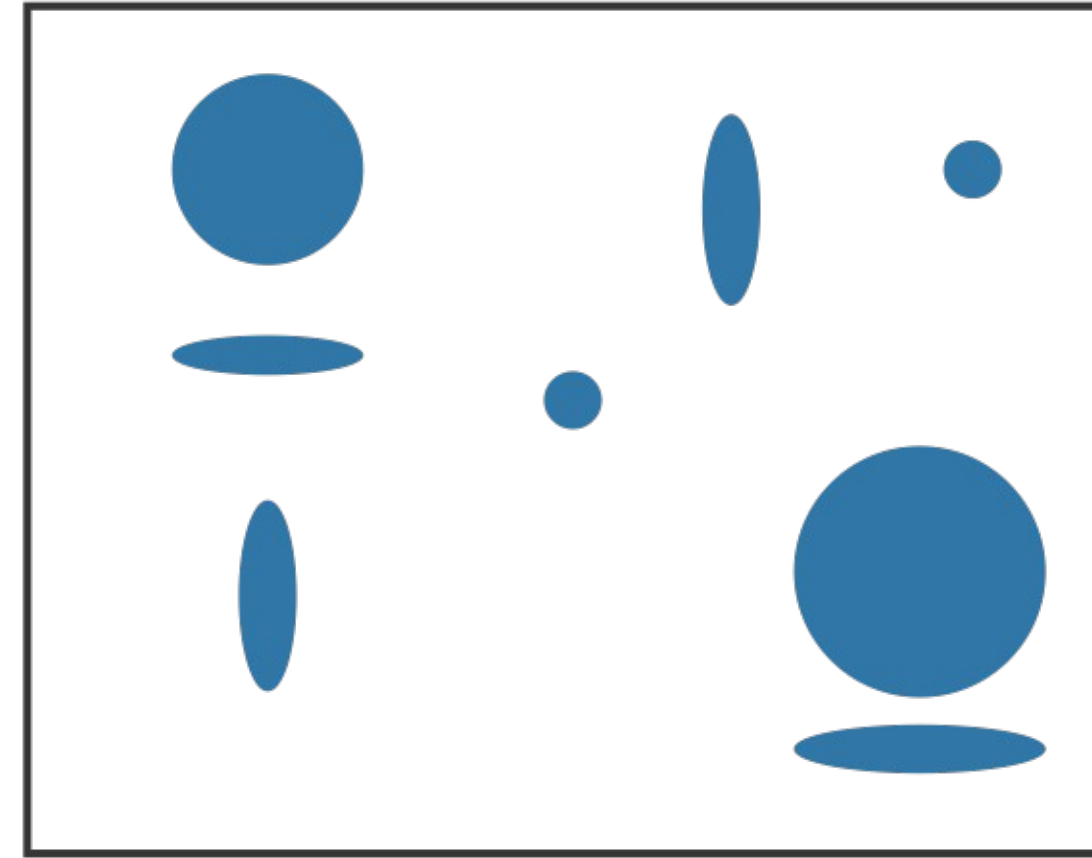
Size
+ Hue (Color)



Some interference

2 groups
each

Width
+ Height

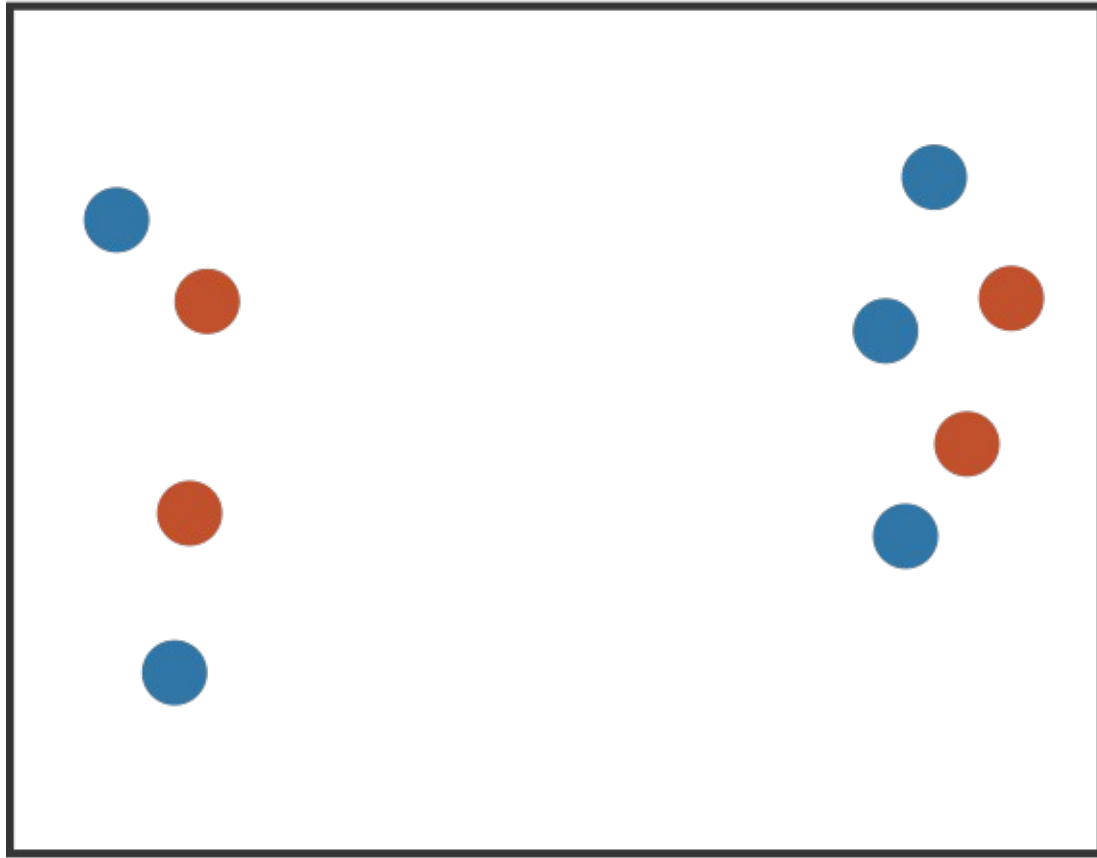


Some/significant
interference

3 groups
total:
integral area

Separability vs. Integrality

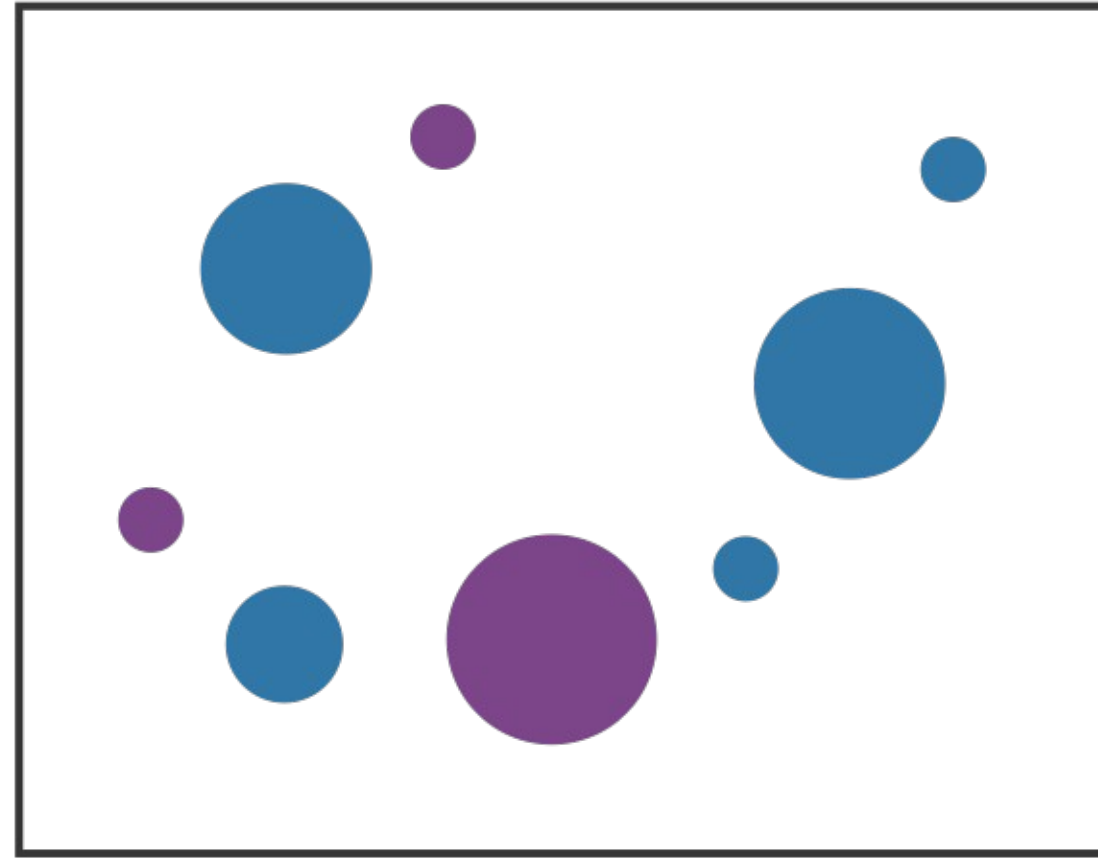
Position
+ Hue (Color)



Fully separable

2 groups
each

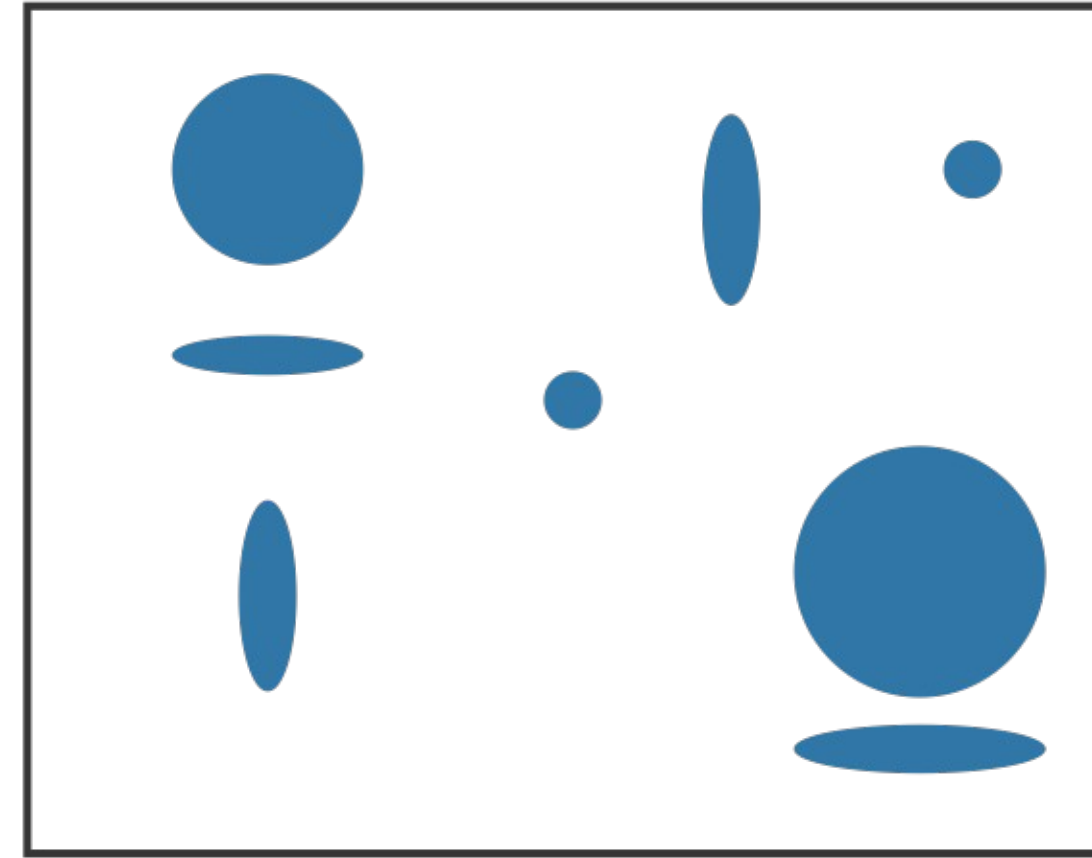
Size
+ Hue (Color)



Some interference

2 groups
each

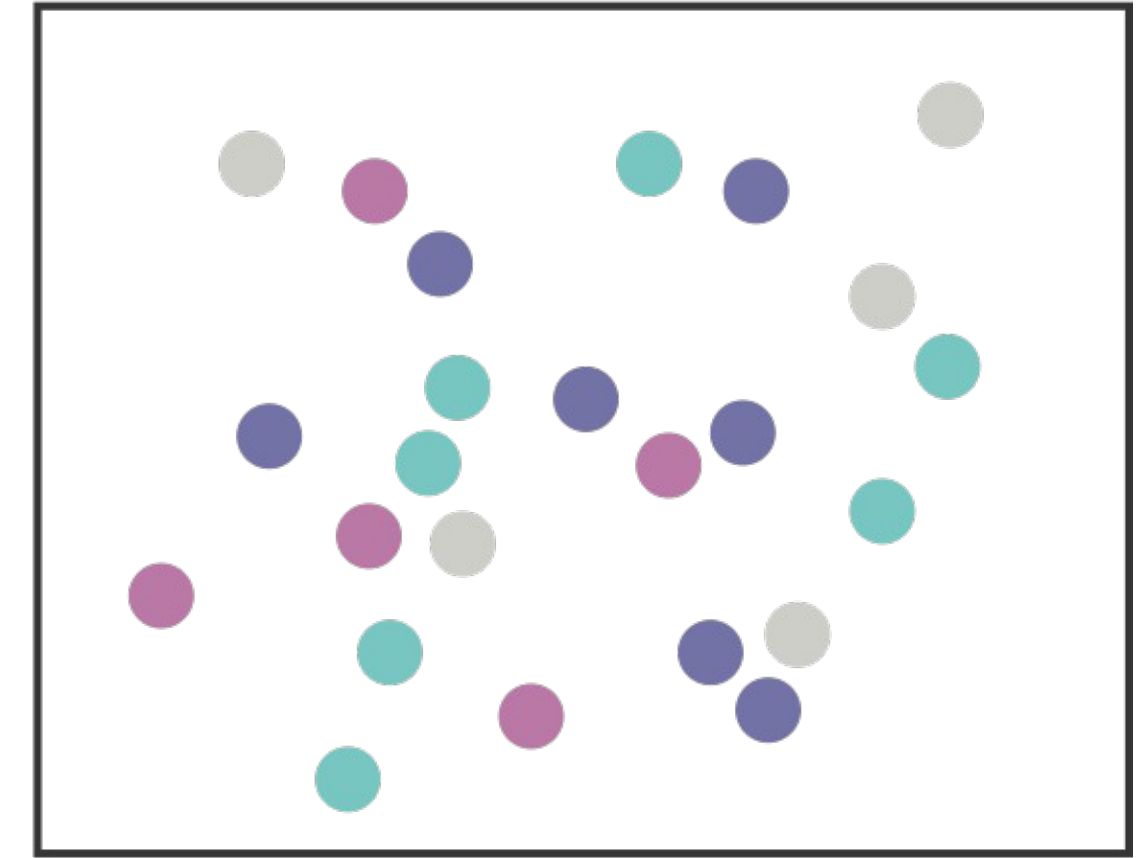
Width
+ Height



Some/significant
interference

3 groups
total:
integral area

Red
+ Green



Major interference

4 groups
total:
integral hue

Popout

Popout

Properties detected by the low-level visual system

very rapid - 200-250 milliseconds

very accurate

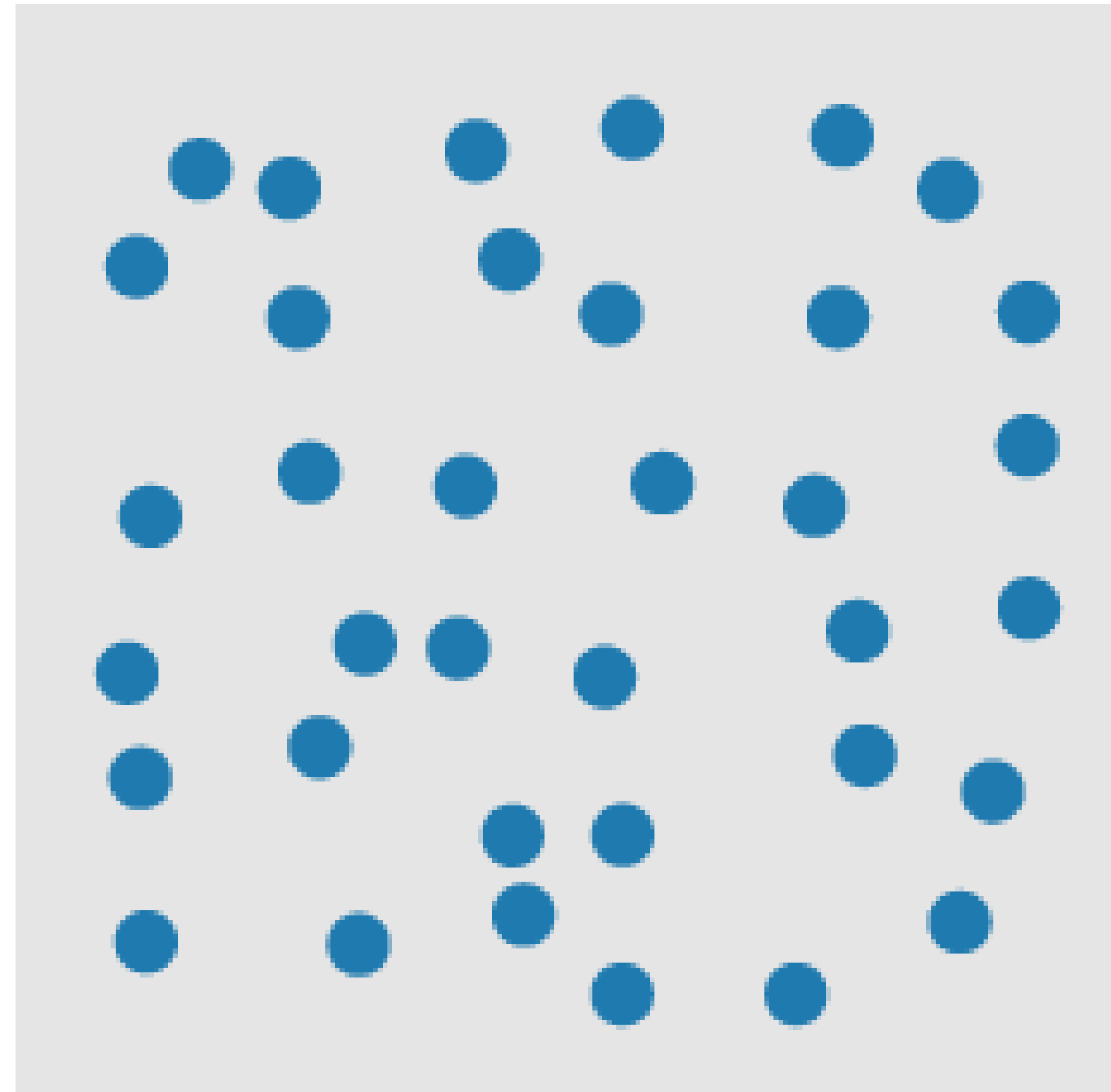
processed in parallel

happens before focused attention -> “pre”-attentive

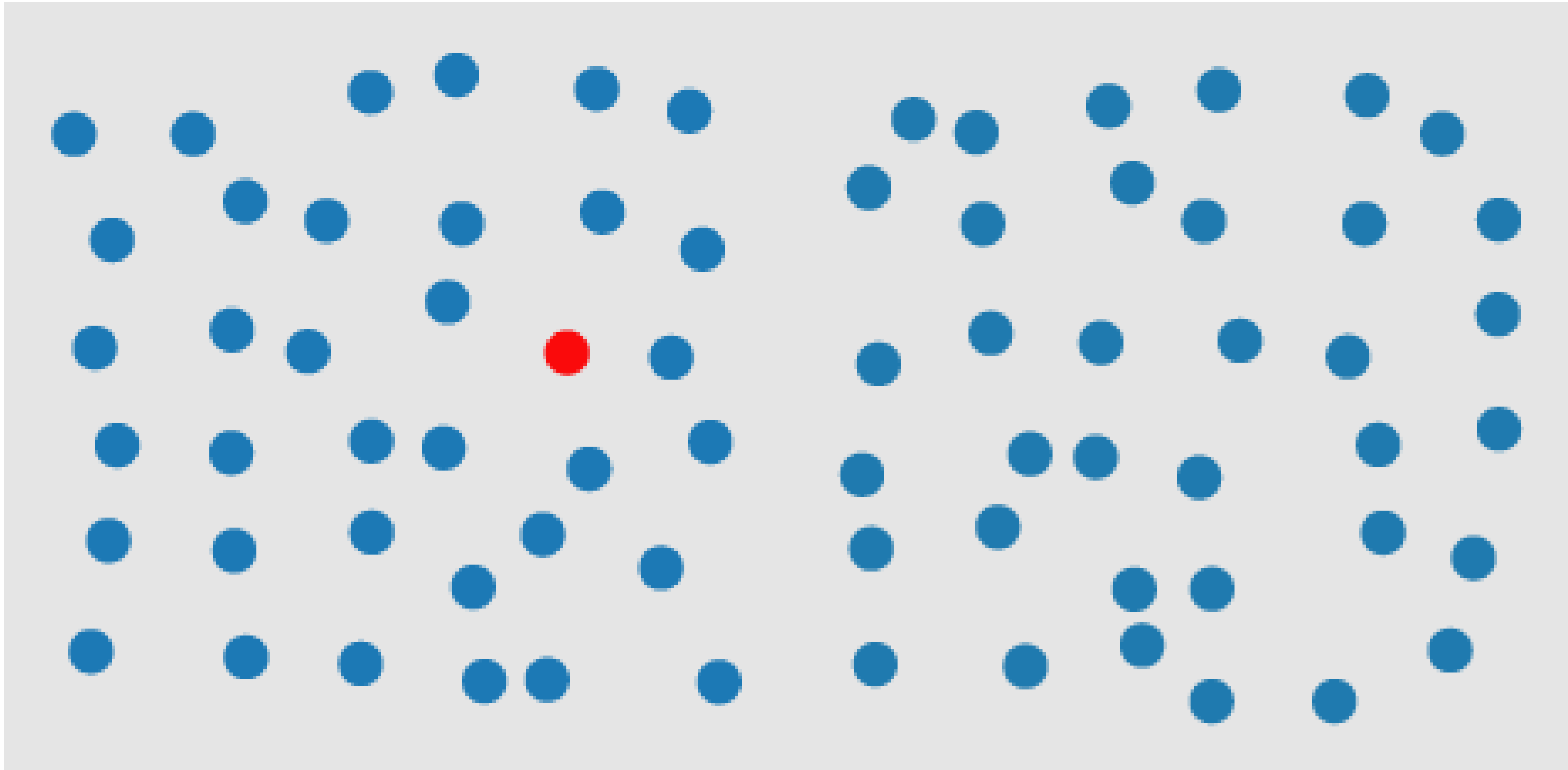
Independent of the number of distractors!

Opposite: sequential search (processed serially)

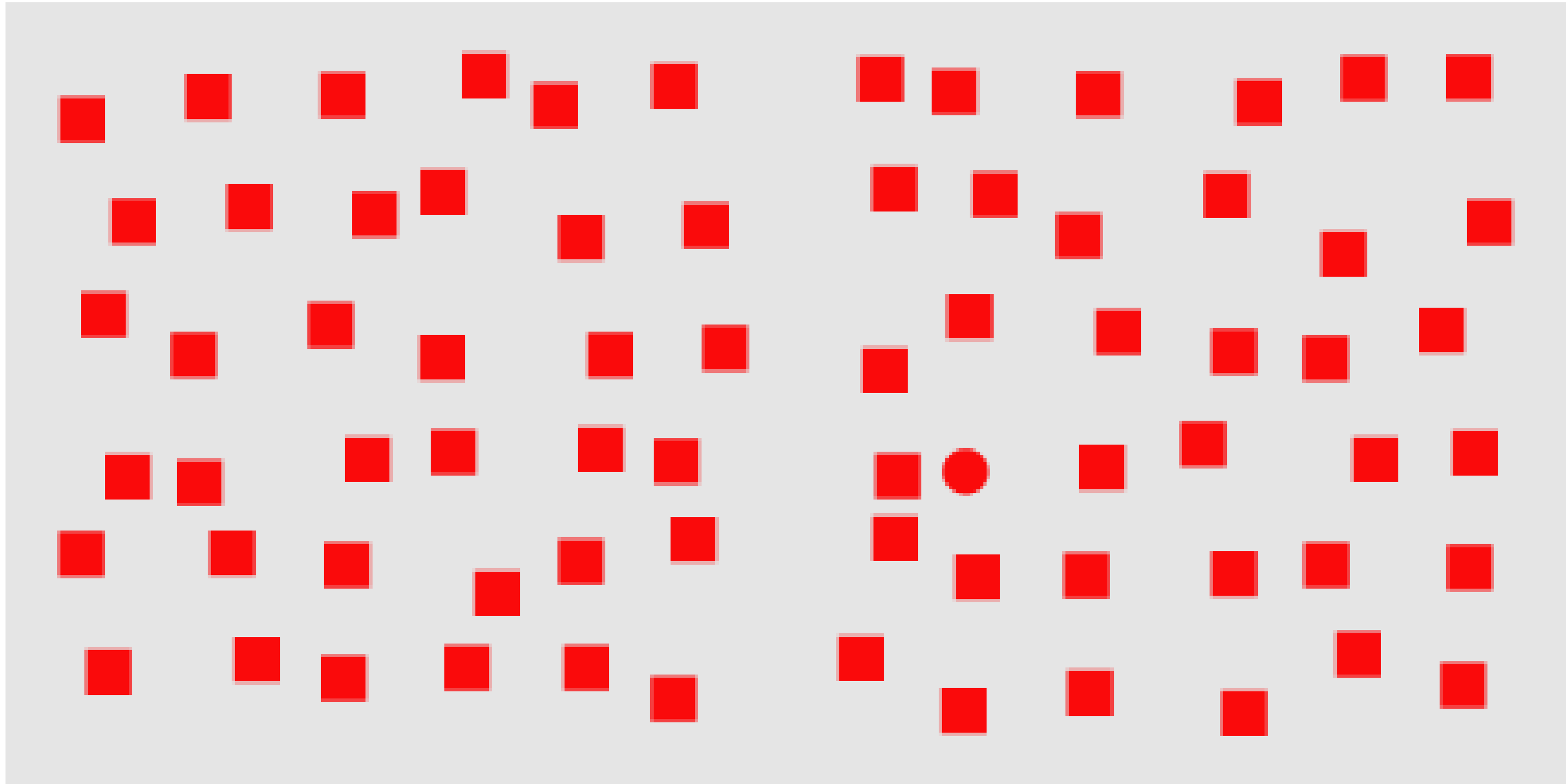
EXPERIMENT

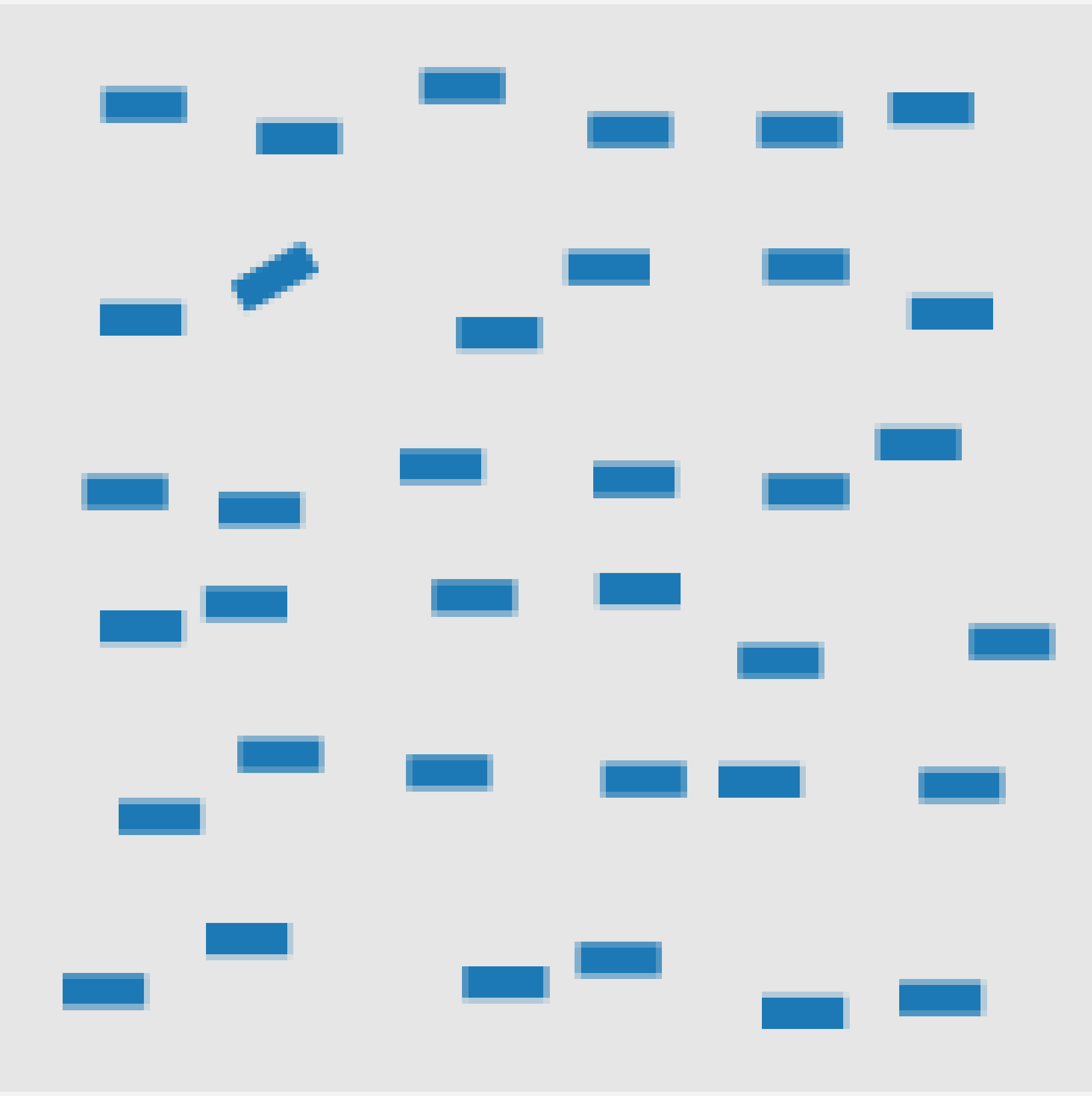


Difference in Hue

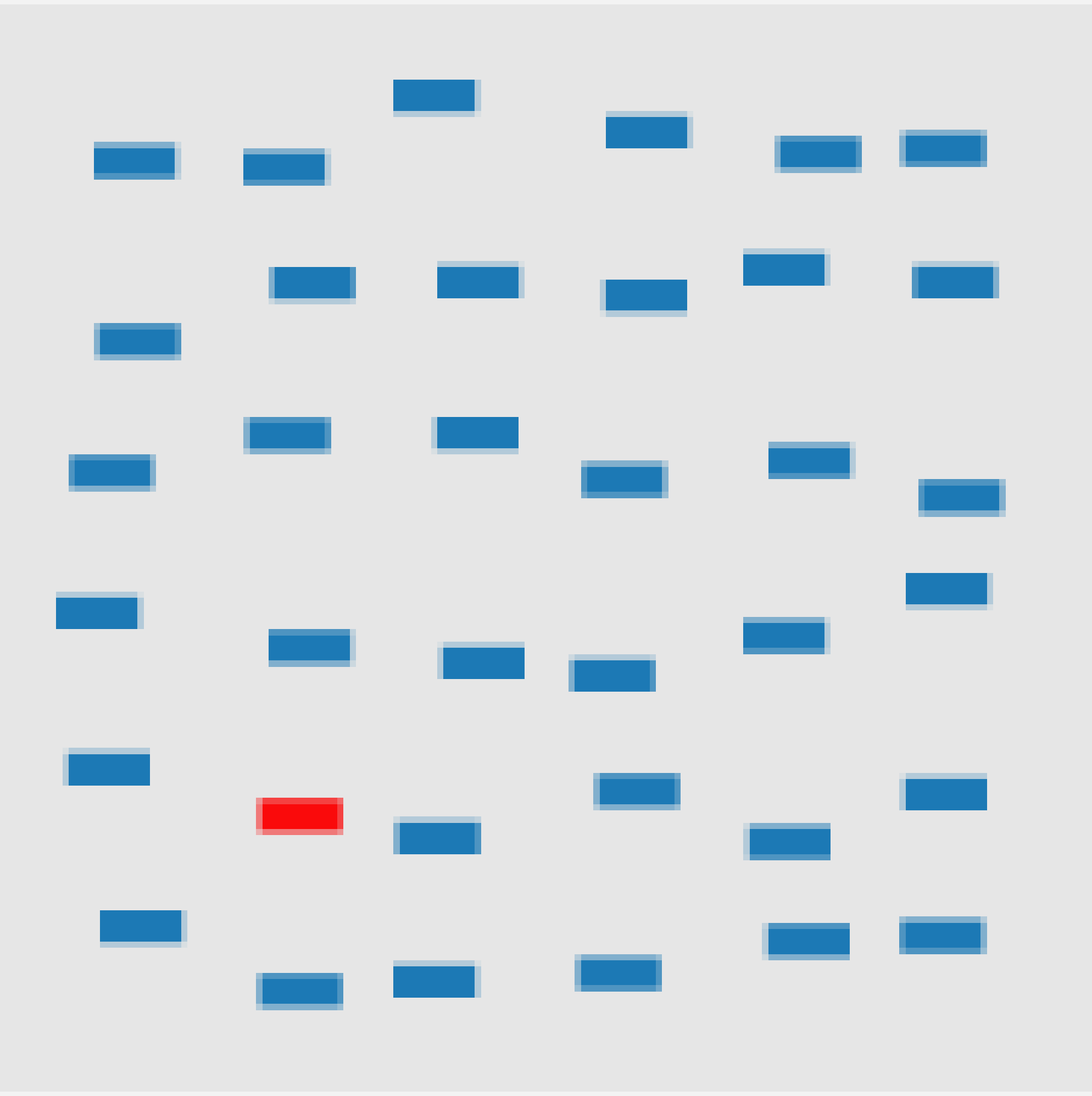


Difference in Curvature / Form

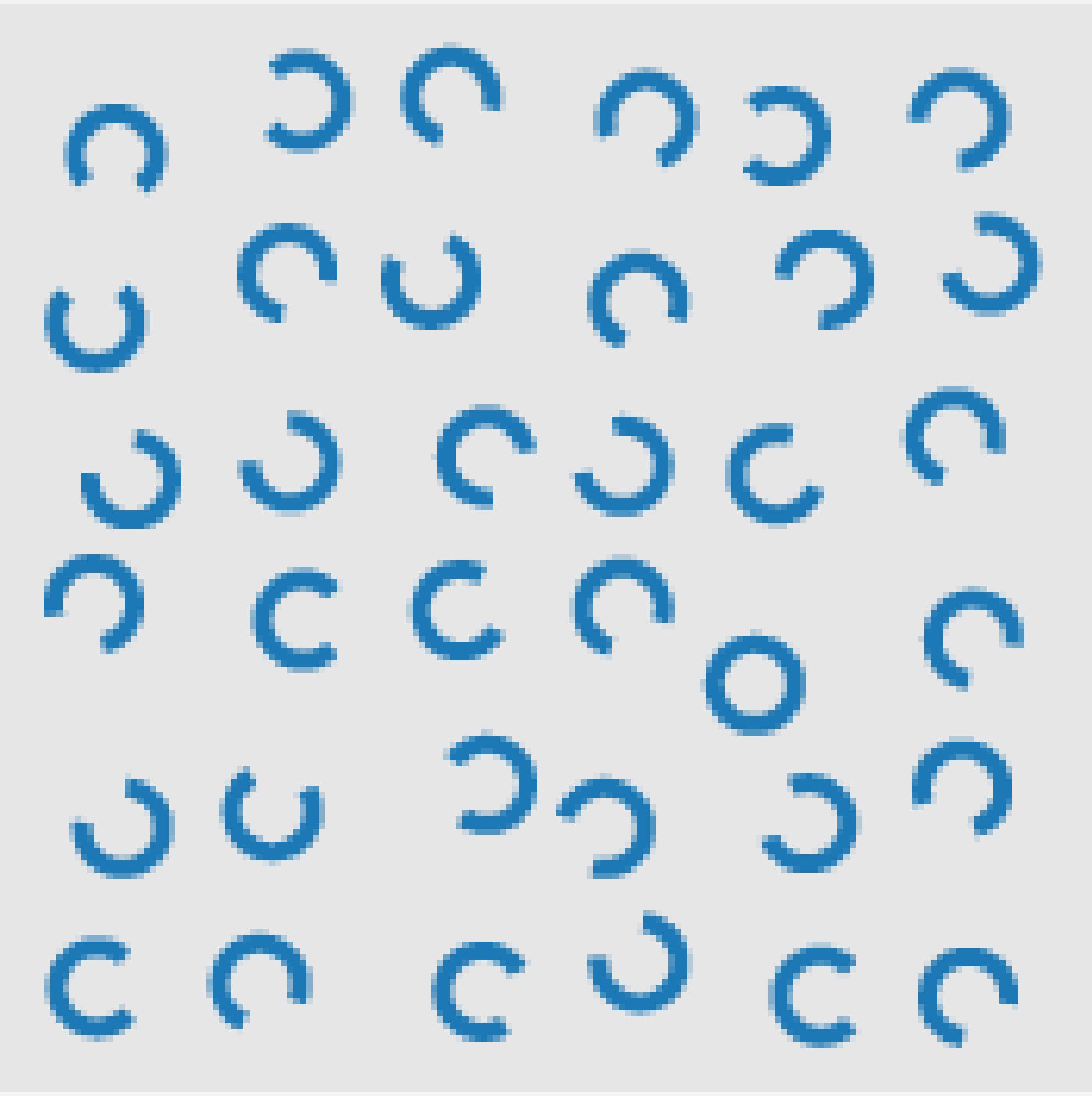




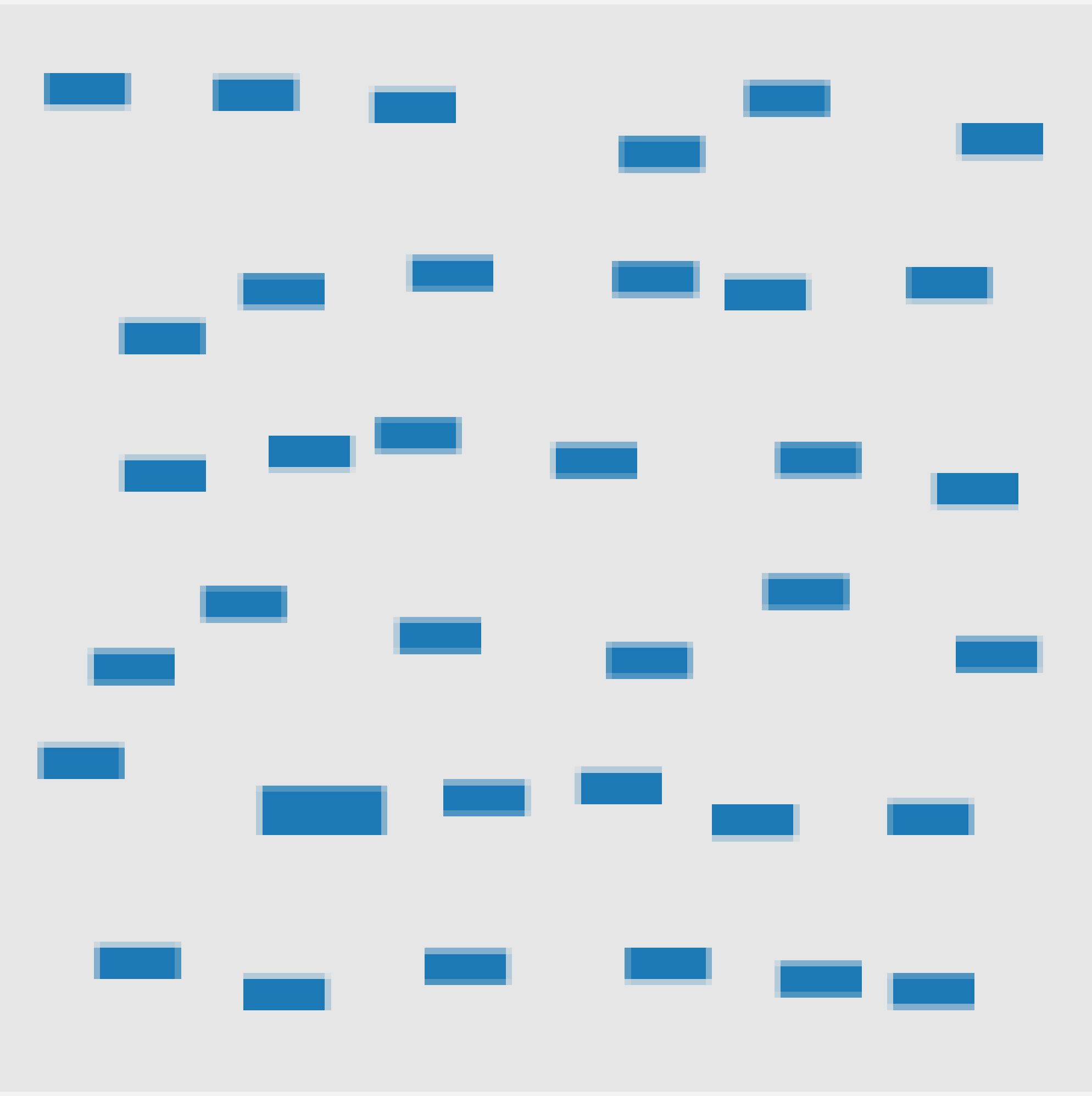
orientation



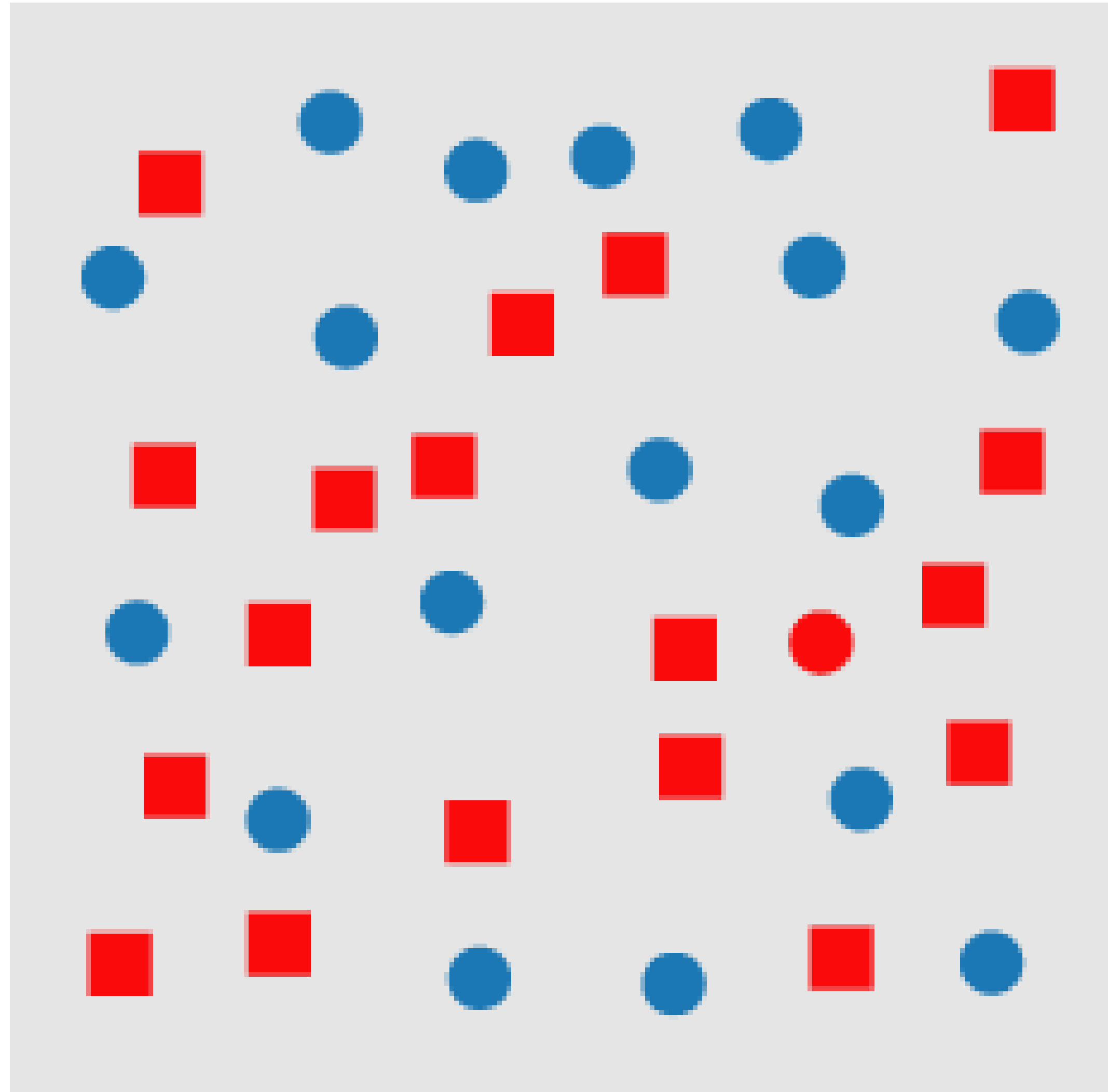
hue



closure

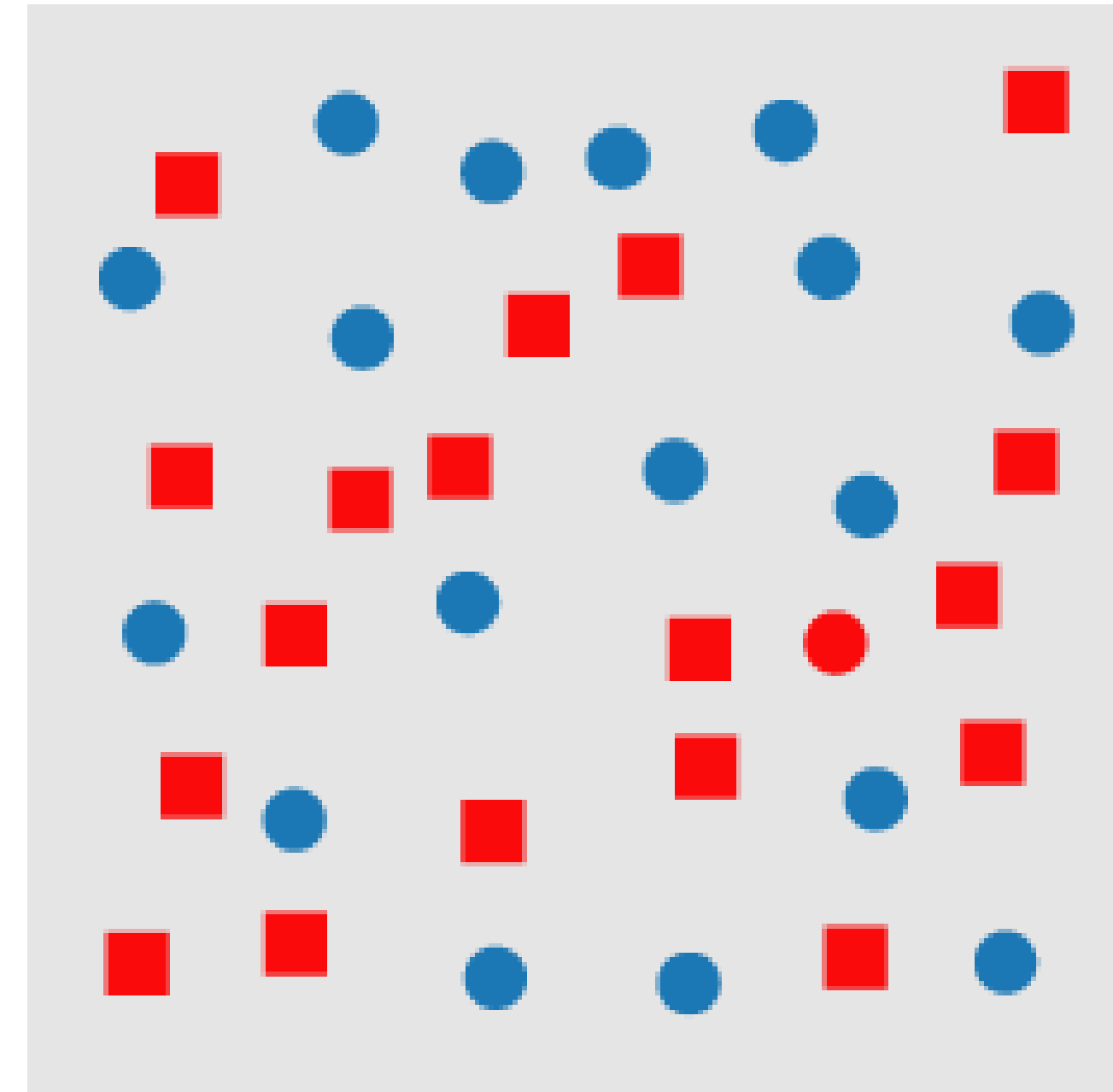
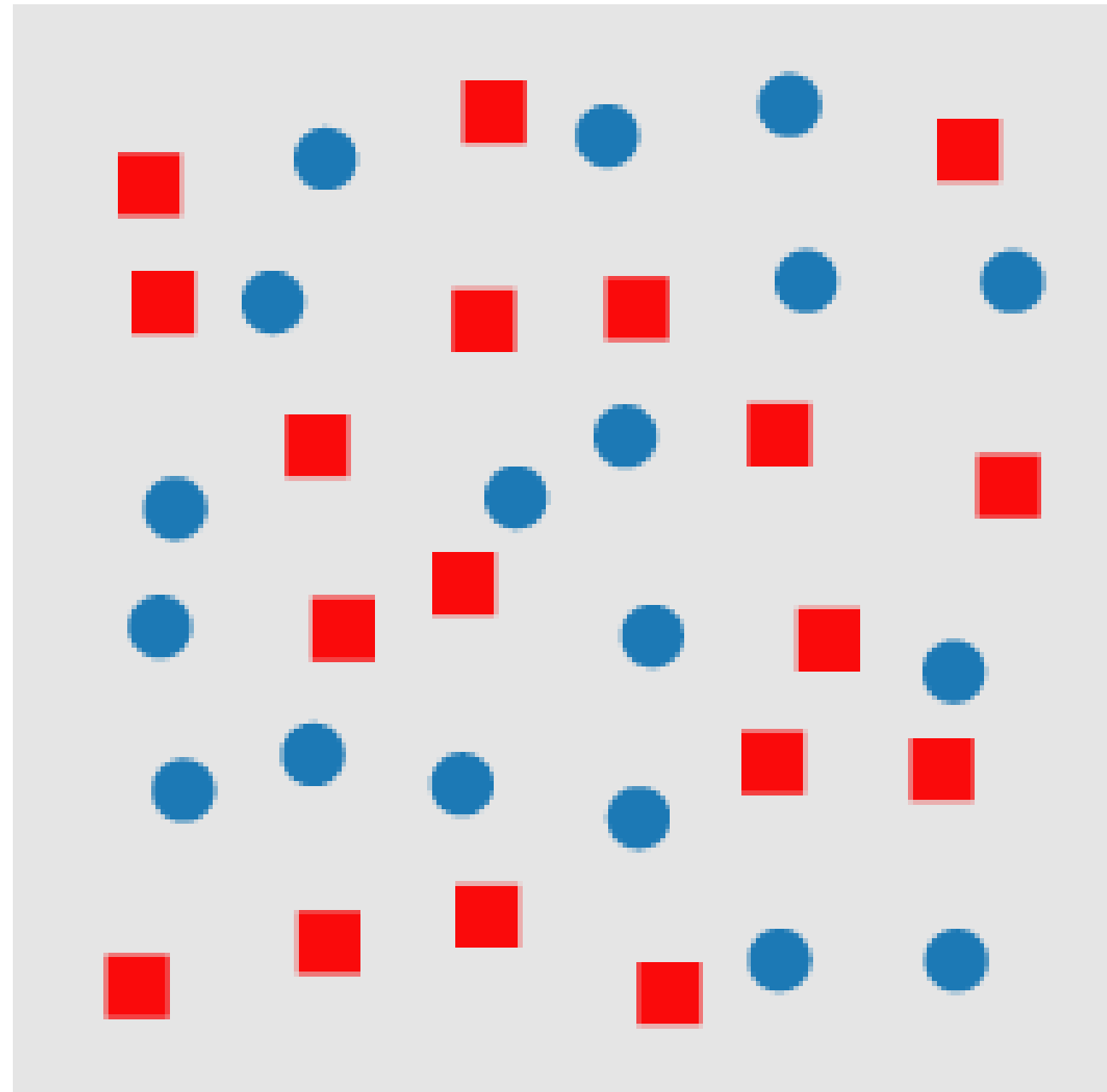


size



?

Not Valid for Combinations

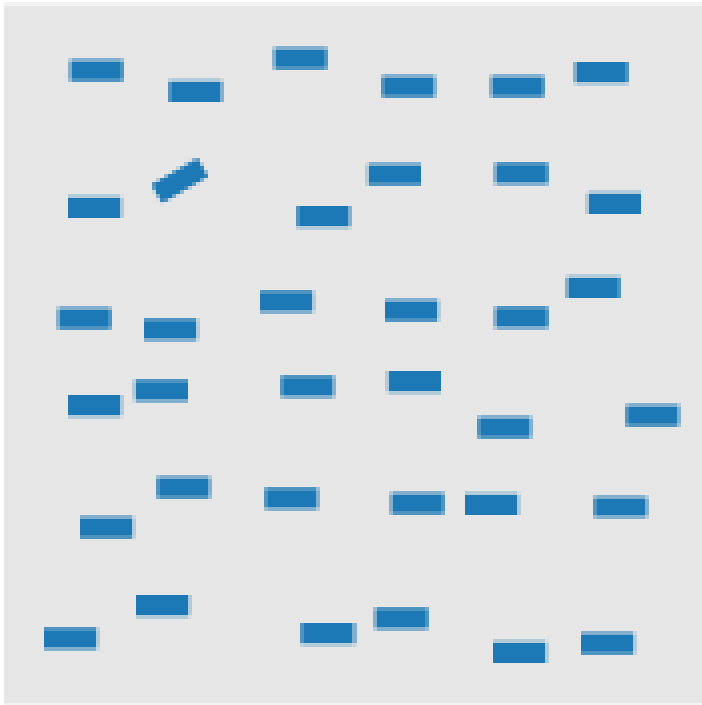


Conjunction Targets – no unique visual property

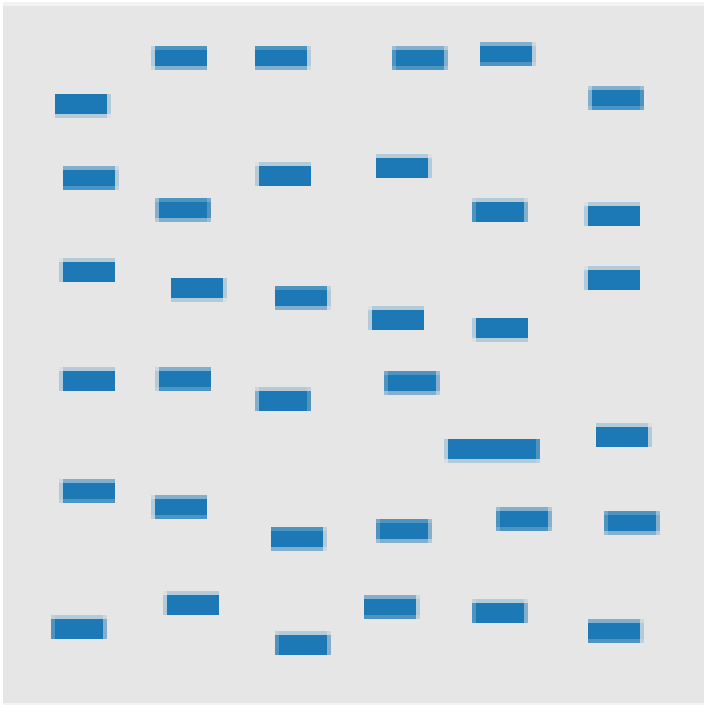
Target: red, circle

Distractor objects have both properties

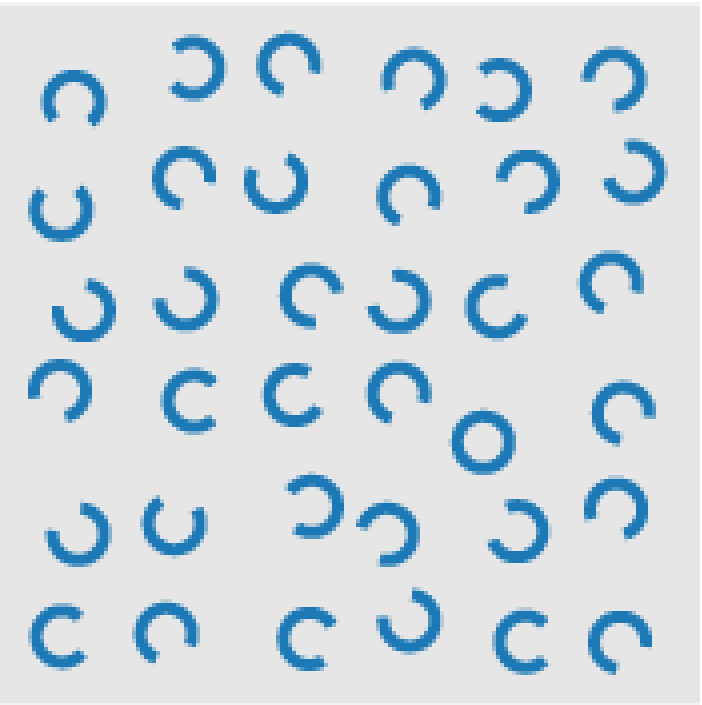
Some Preattentive Properties



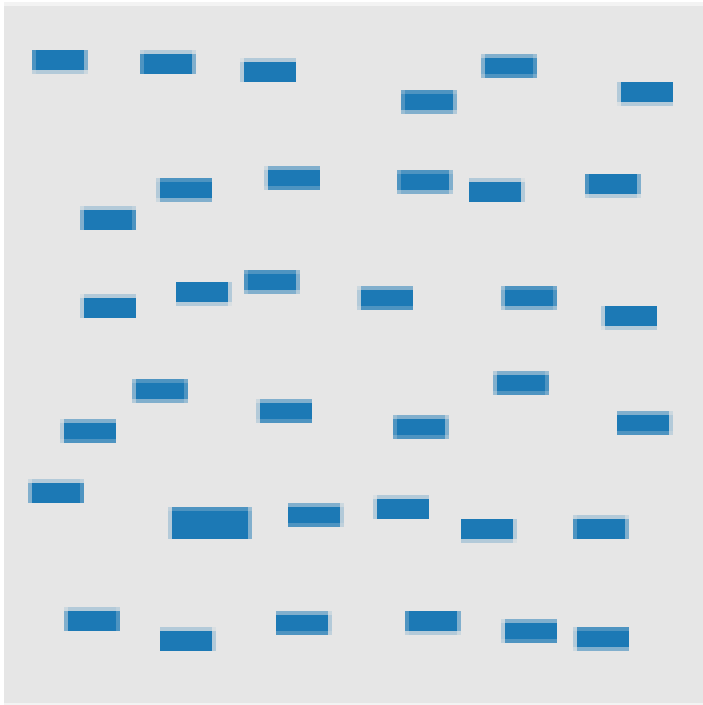
orientation



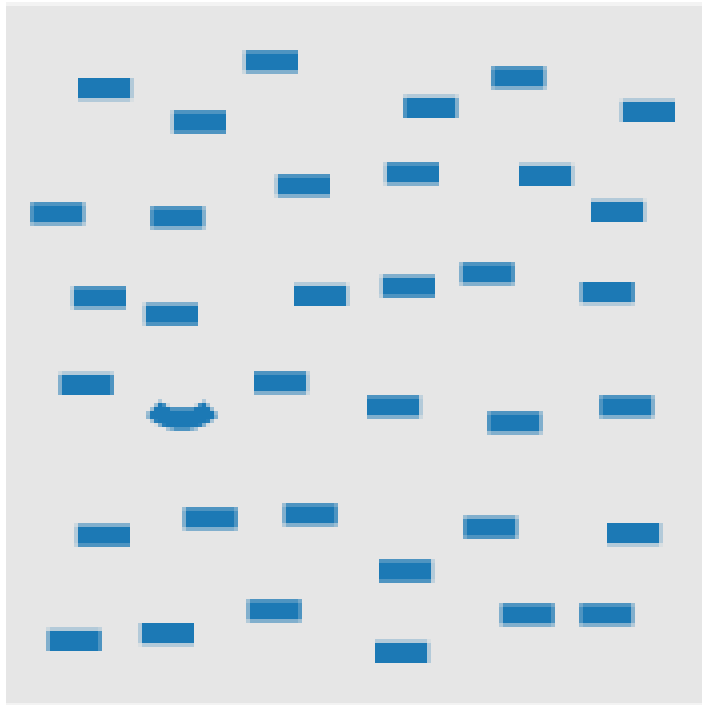
length



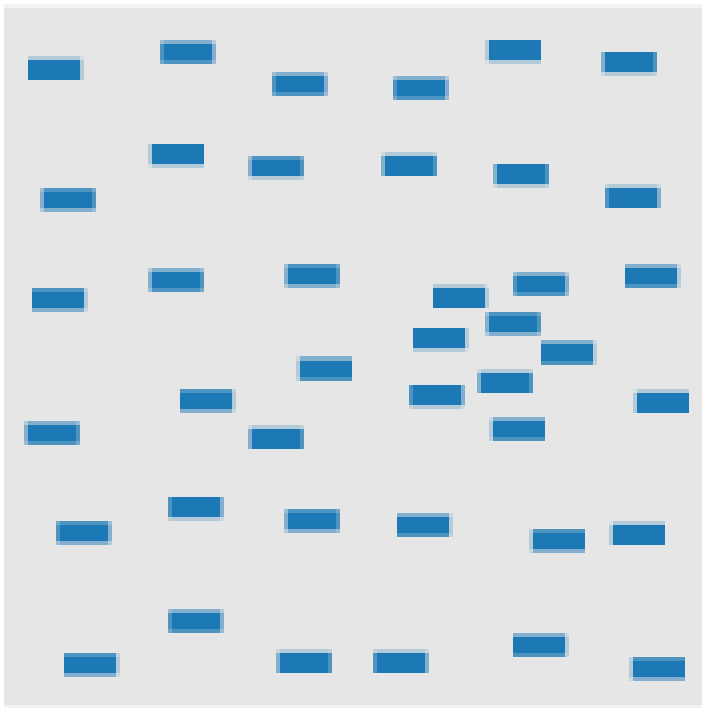
closure



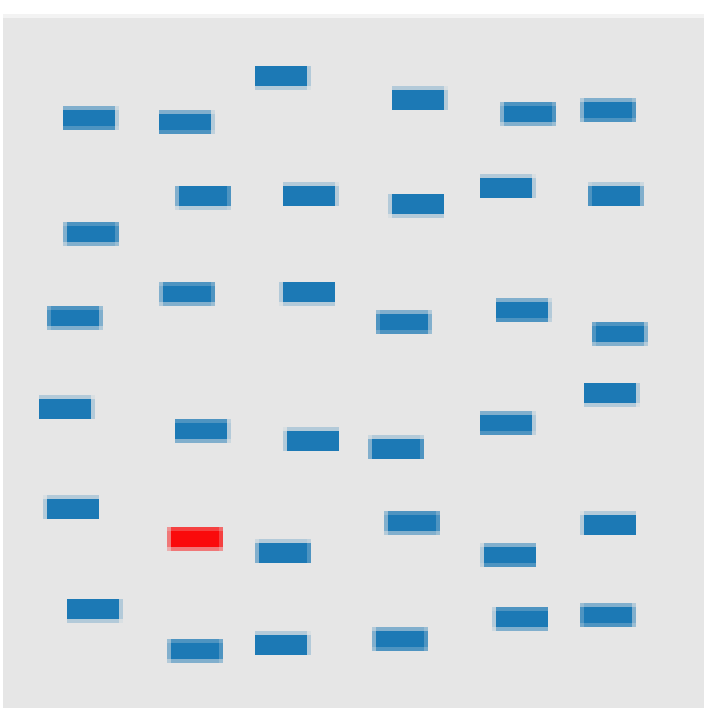
size



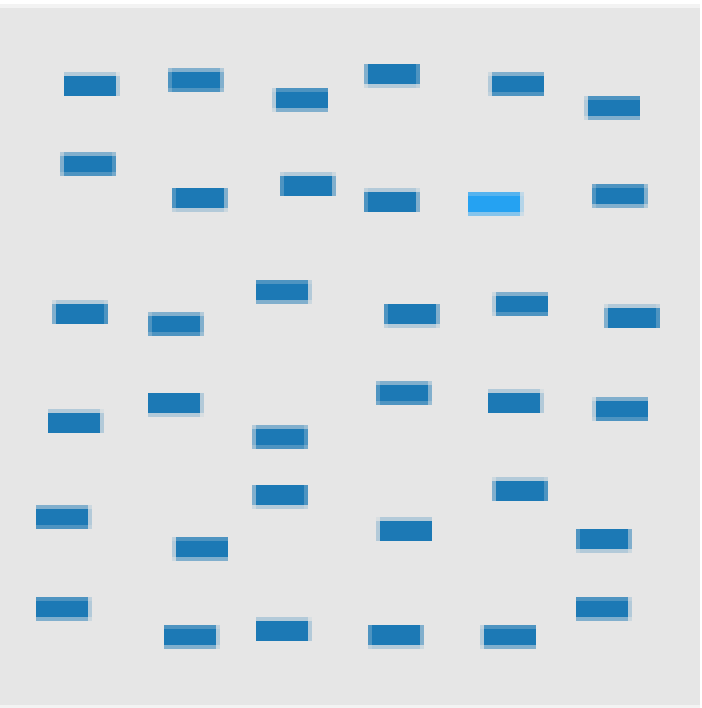
curvature



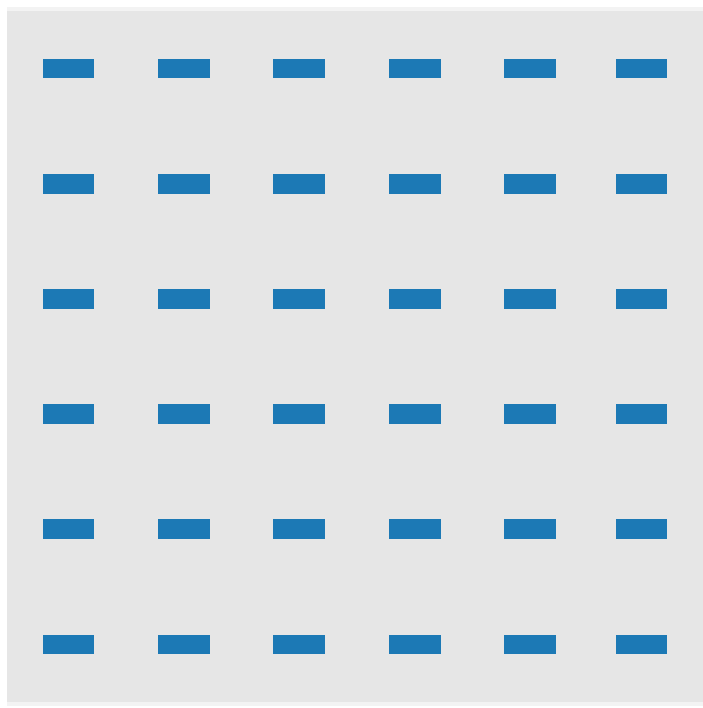
density



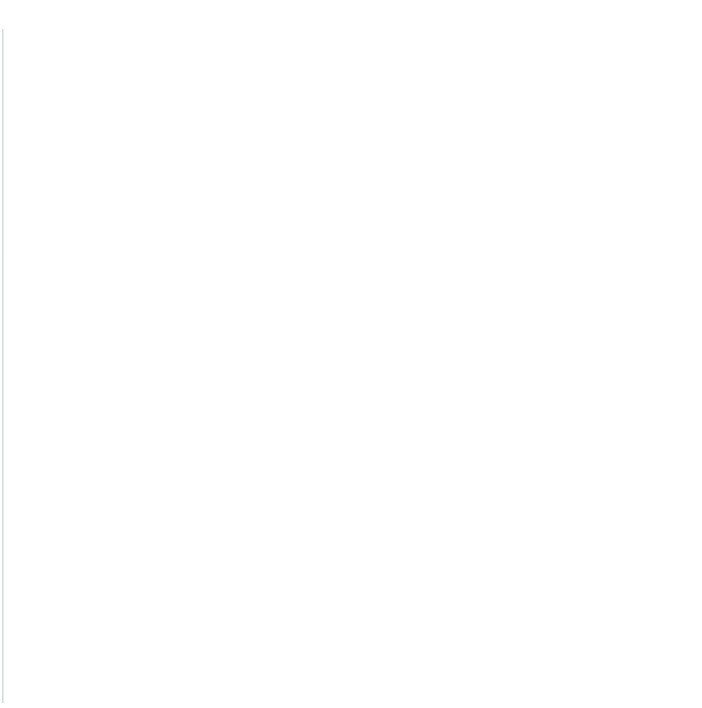
hue



hue



flicker

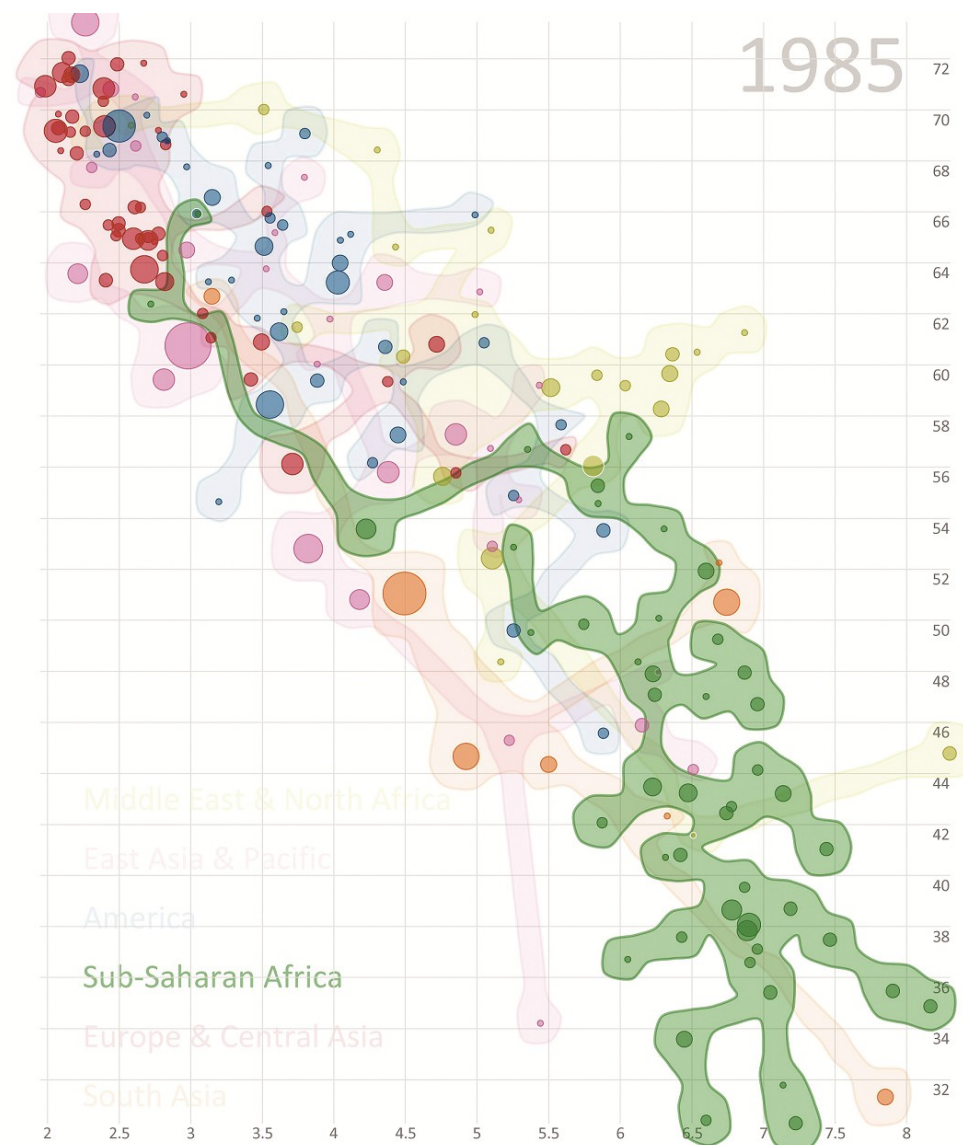
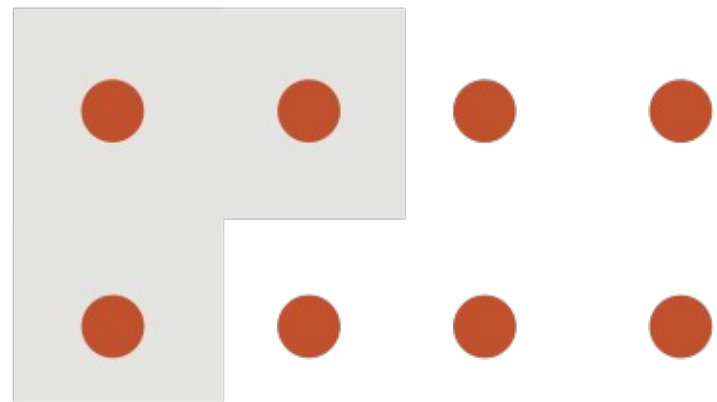


direction of motion

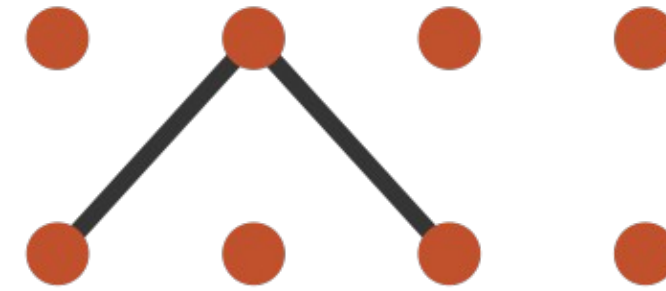
Grouping

Marks for links

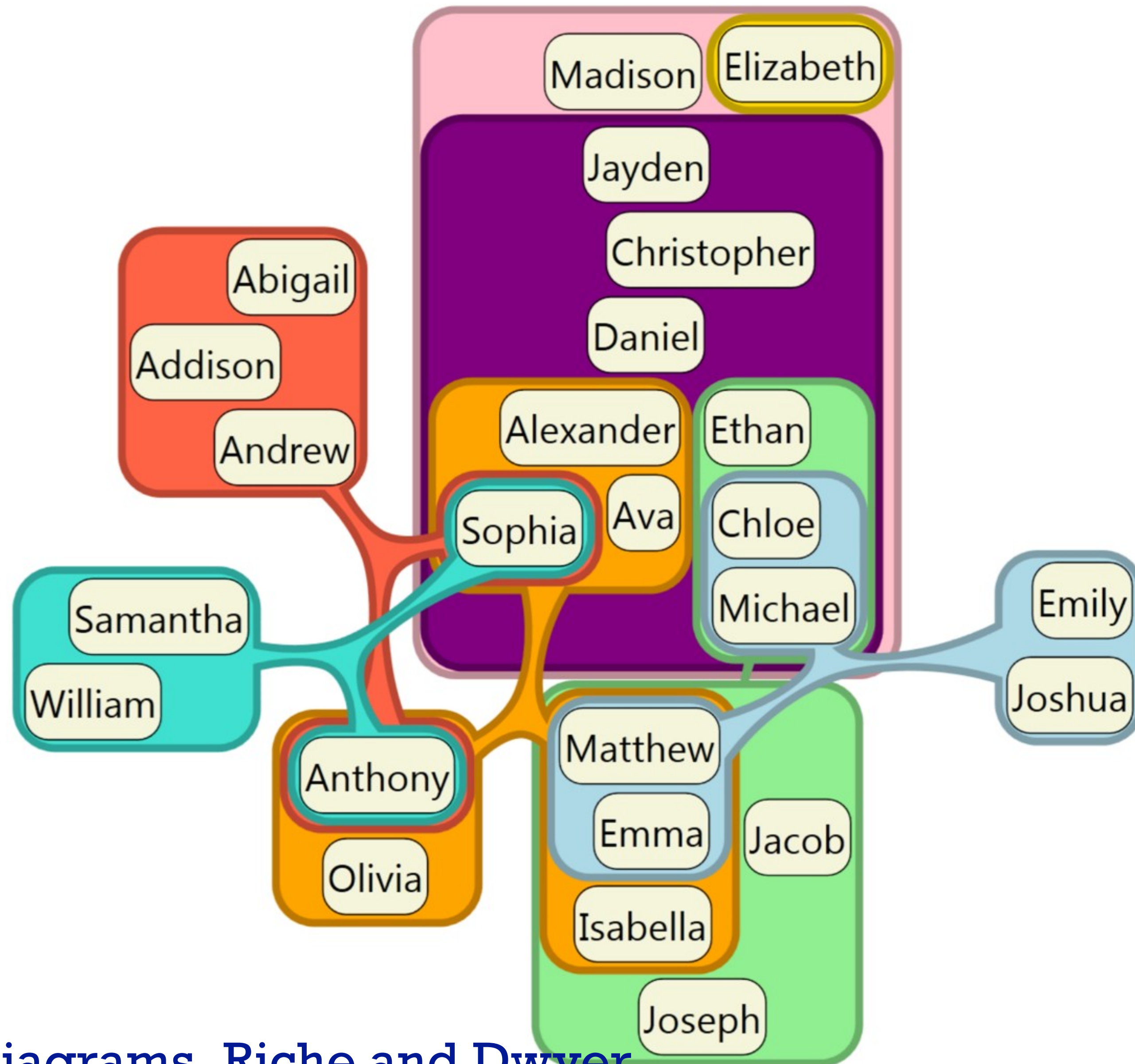
➔ Containment



➔ Connection



Containment can be nested



[Untangling Euler Diagrams, Riche and Dwyer,

00101

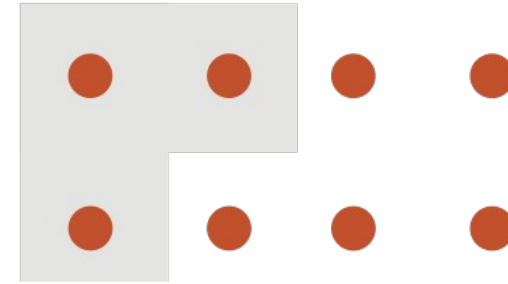
Grouping

- containment
- connection

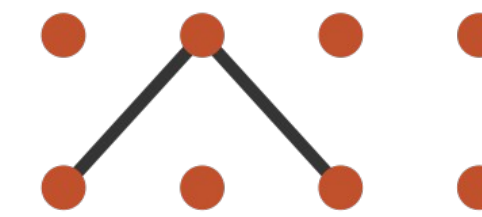
- **proximity**
 - same spatial region
- **similarity**
 - same values as other categorical channels

Marks as Links

➔ Containment



➔ Connection



➔ Identity Channels: Categorical Attributes

Spatial region



Color hue



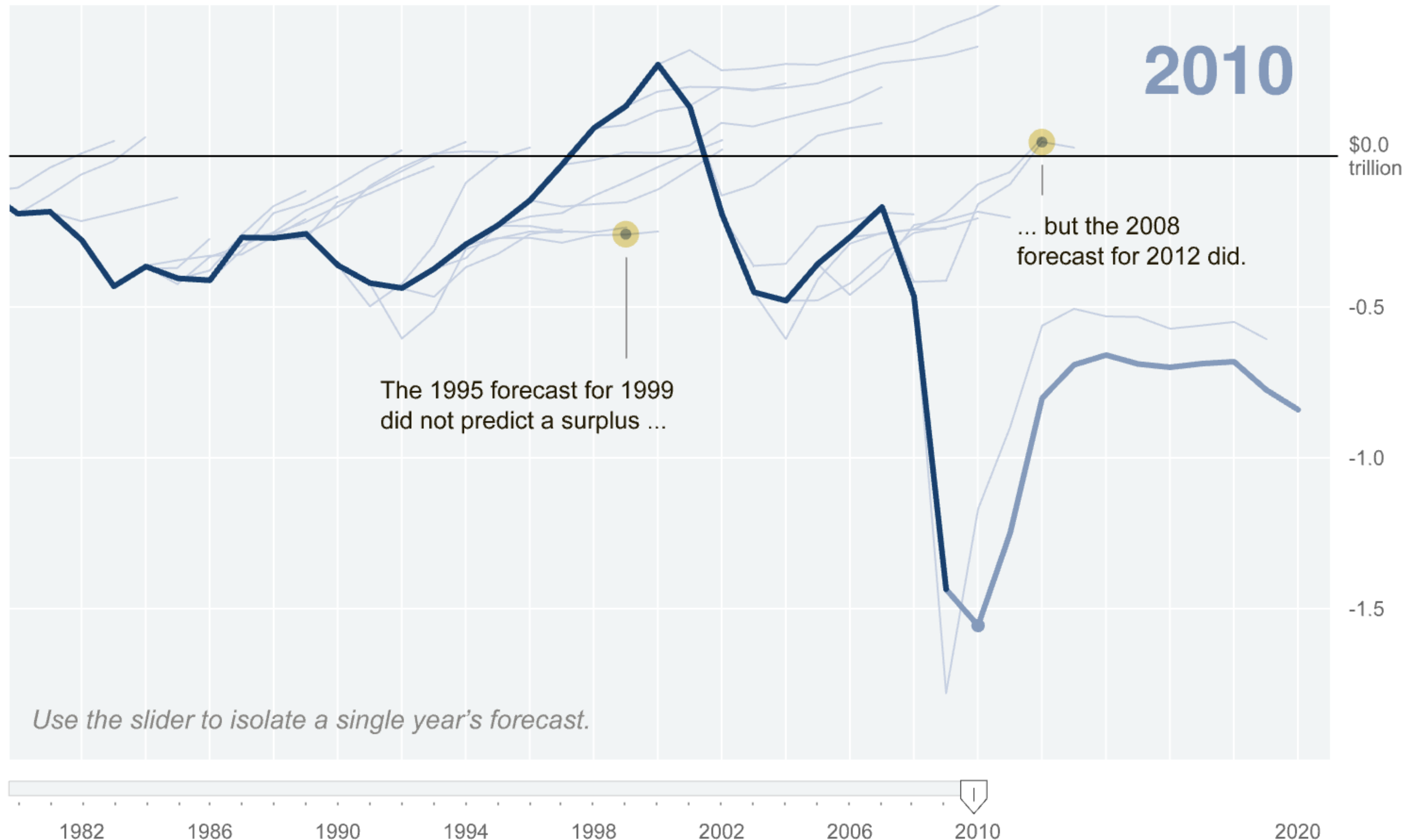
Motion



Shape

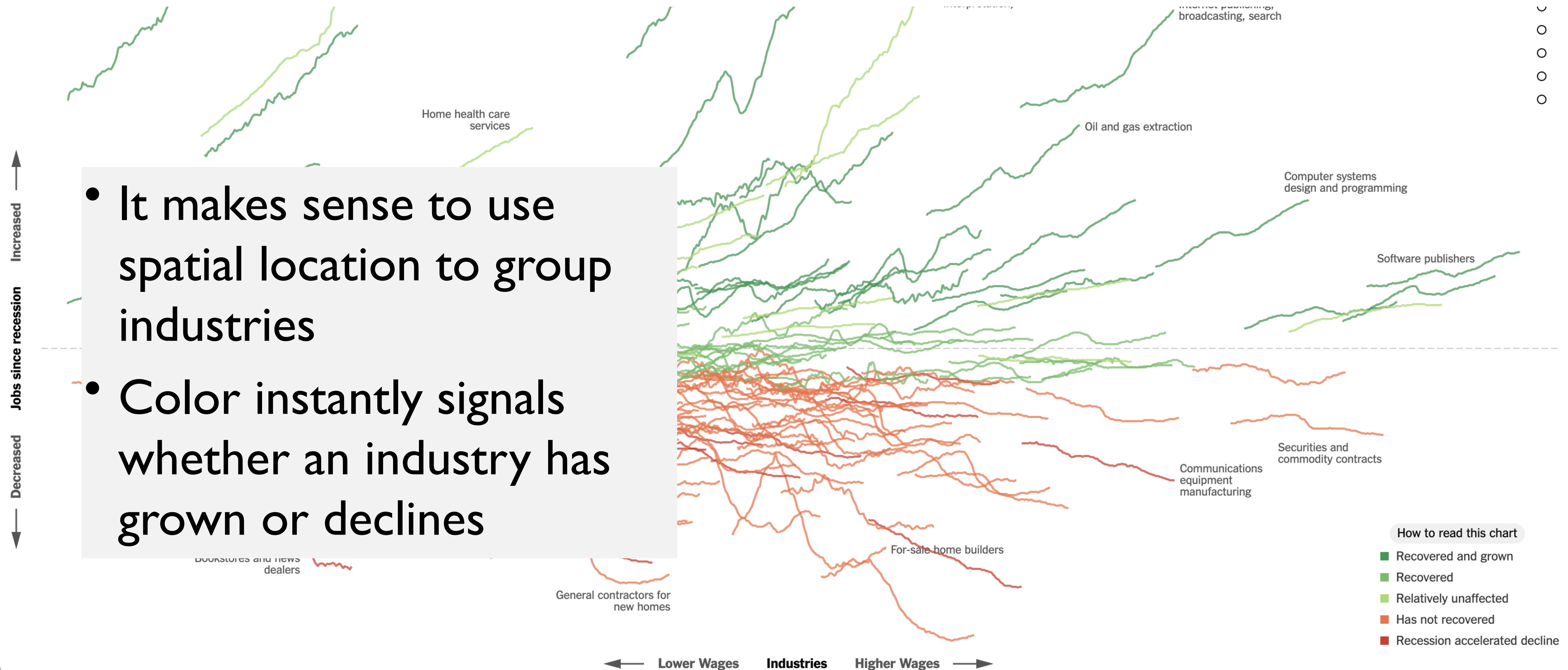


Using grouping for emphasis vs background



- By using similarity for the projections (light blue), they visually group together. The dark blue pop-out is what actually happened.

Proximity and similarity to see patterns in groups across sectors



Map Key

One dot = 200 people

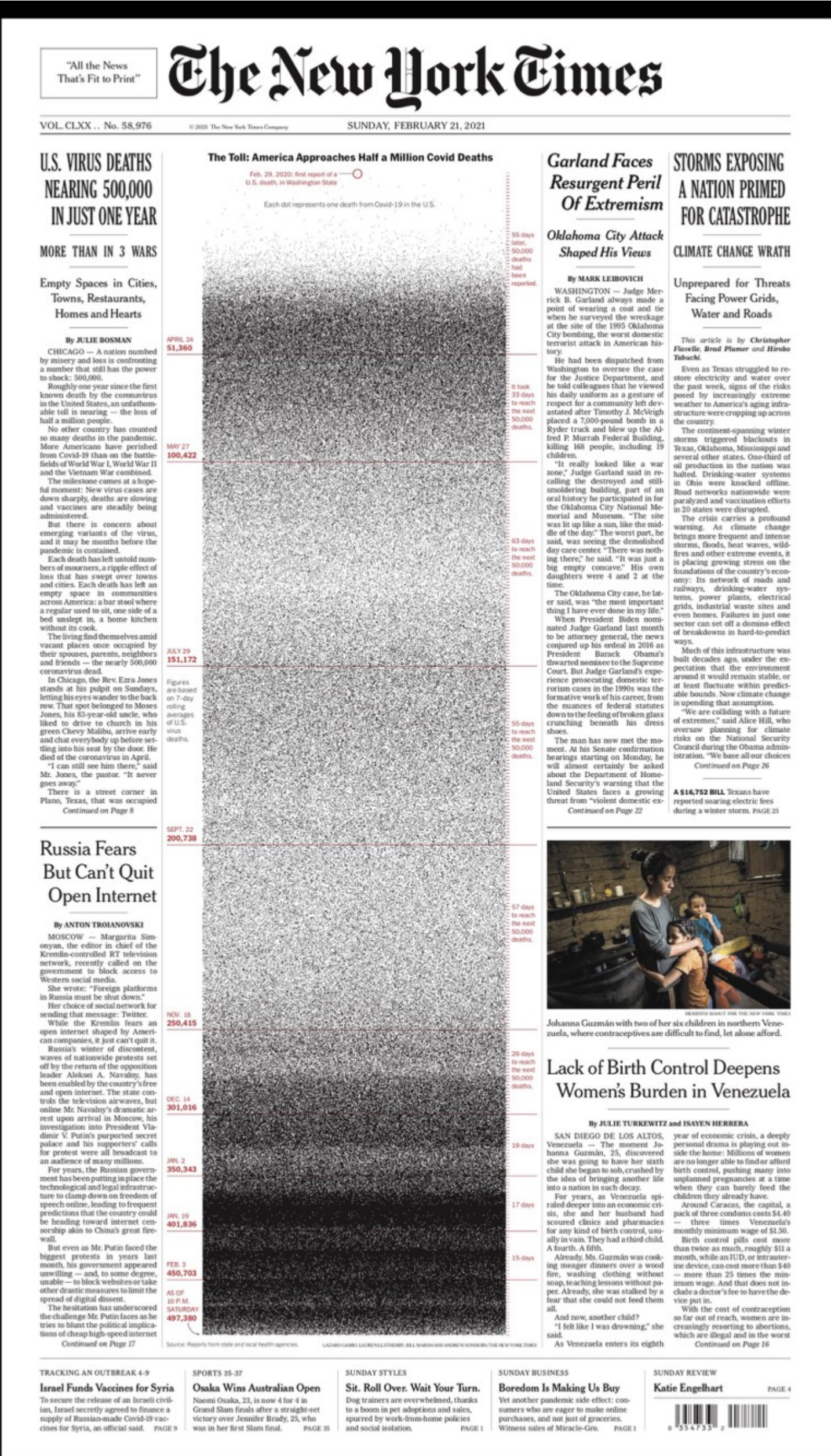
CENSUS GROUP

- Black
- Hispanic
- Asian
- White
- Others

[https://
www.nytimes.com/
interactive/
2015/07/08/us/
census-race-
map.html](https://www.nytimes.com/interactive/2015/07/08/us/census-race-map.html)

Proximity naturally shows patterns in the data: waves of COVID

- Times when COVID deaths surged are not emphasized through additional graphical marks, but they are evident purely from the proximity of points/**grouping** of the data



<https://x.com/nytimes/status/1363498989652426754/photo/1>

Preattentive processing: data viz

Can be used to draw attention to areas of interest

Can be used to express similarity/group memberships

Visual features must be carefully designed

Conjunctions must be avoided

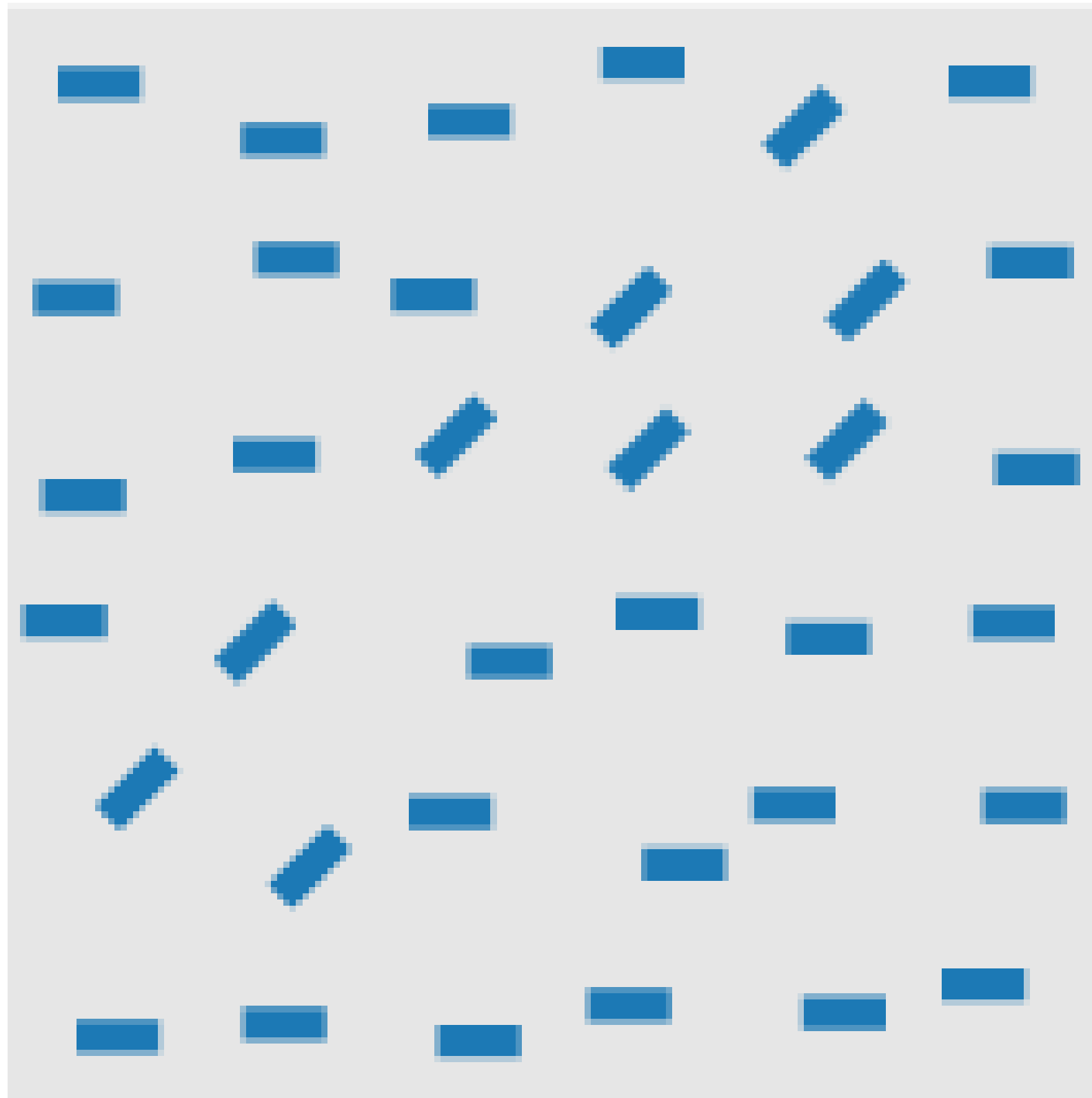
Remember that you won't always be there to speak in front of your visualization **to guide people's attention.**

Takeaways

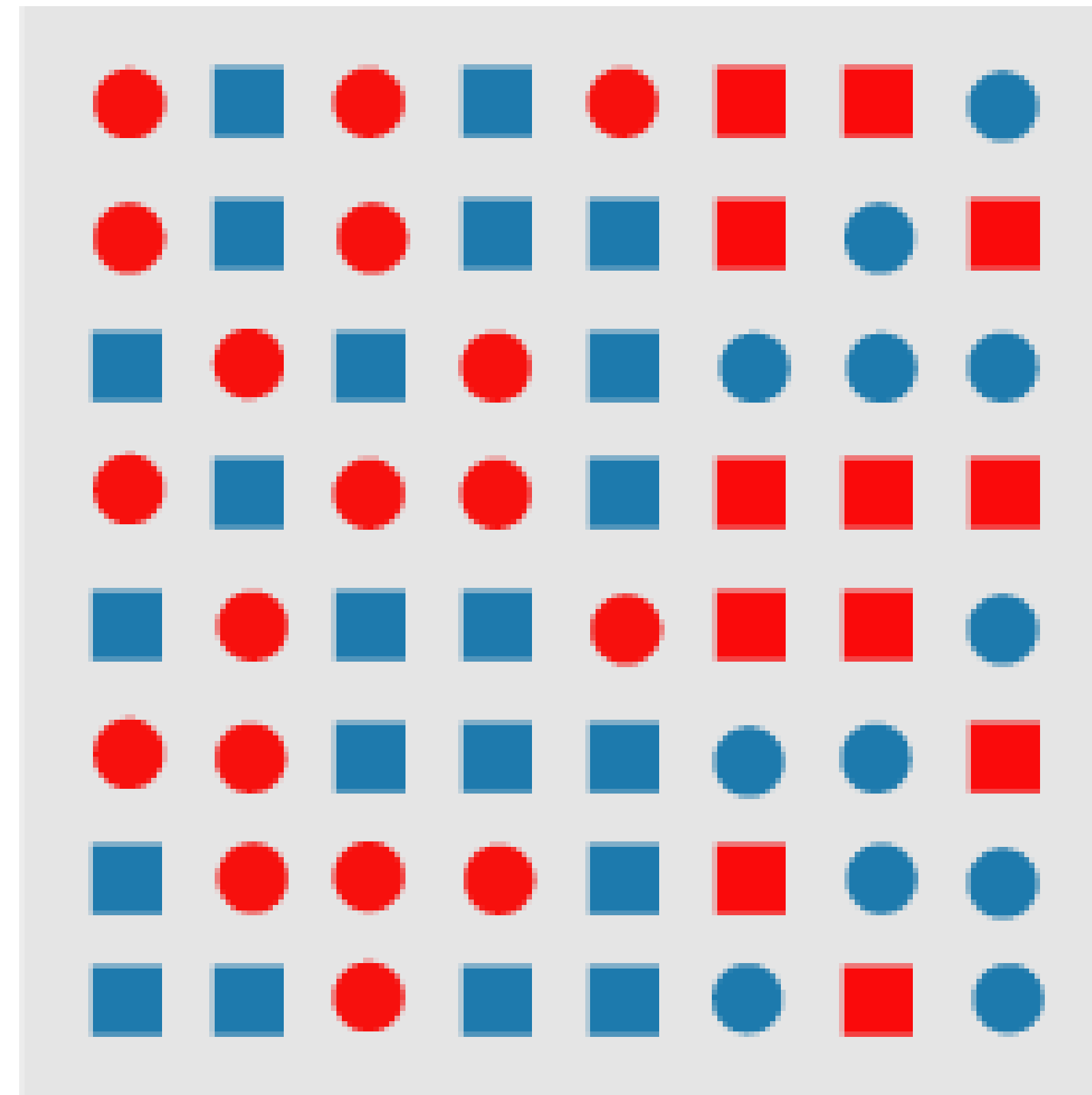
- We've learned other criteria for making effective visualizations: **discriminability**, **separability**, ability to provide **popout**, and ability to provide perceptual **groupings**
- WITH A PARTNER, in the notes section of our Guidelines google doc: what would you suggest to a friend who is building a visualization, based on what you learned today? Add **TWO suggestions**.
- I will compile them into the main document

Tasks

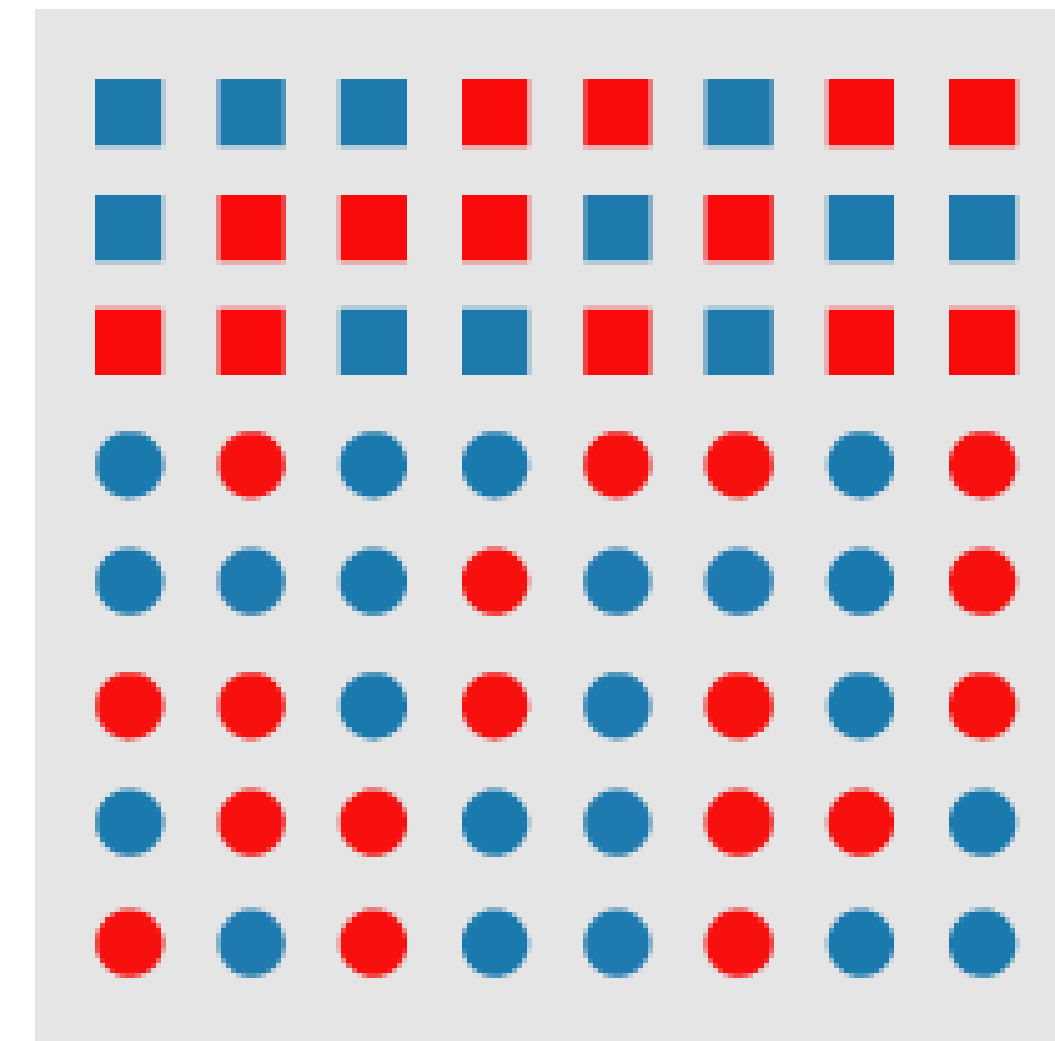
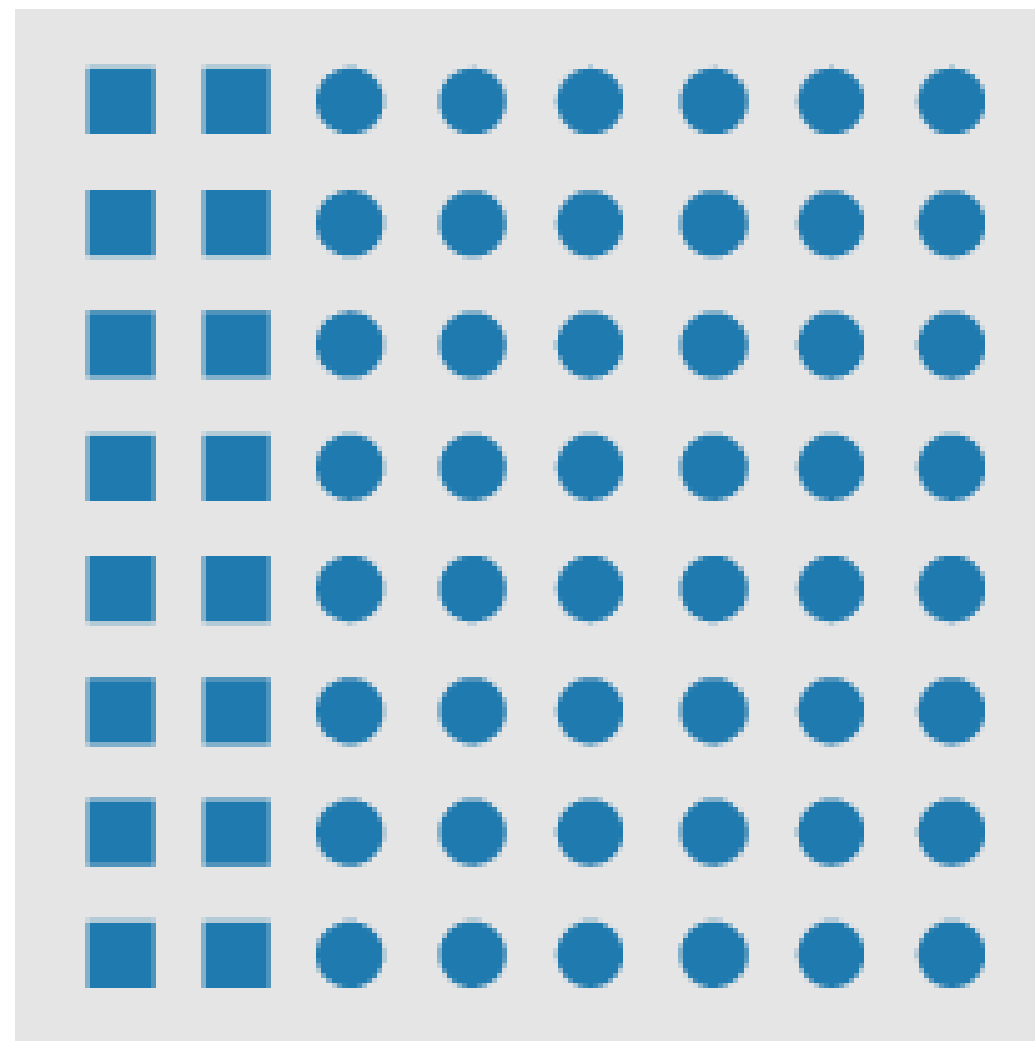
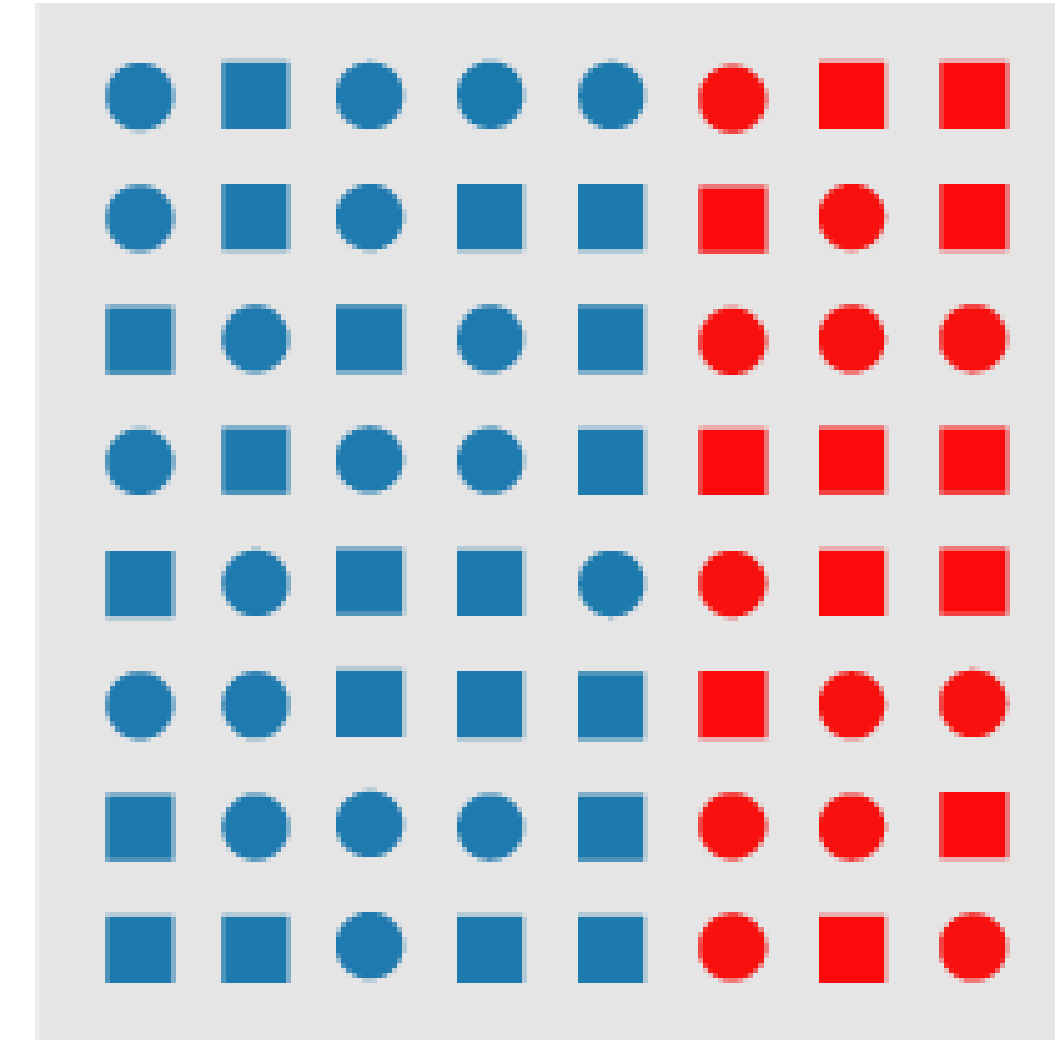
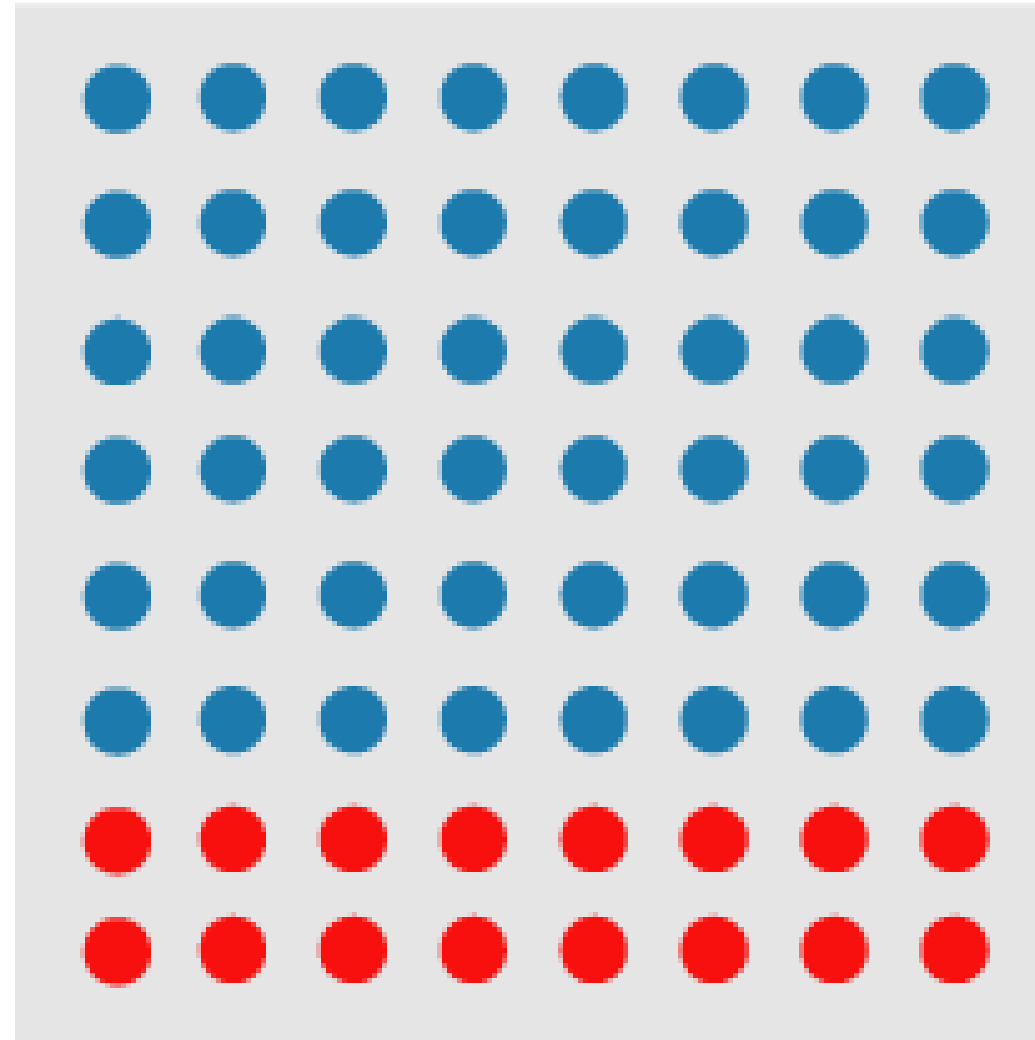
Number Estimation



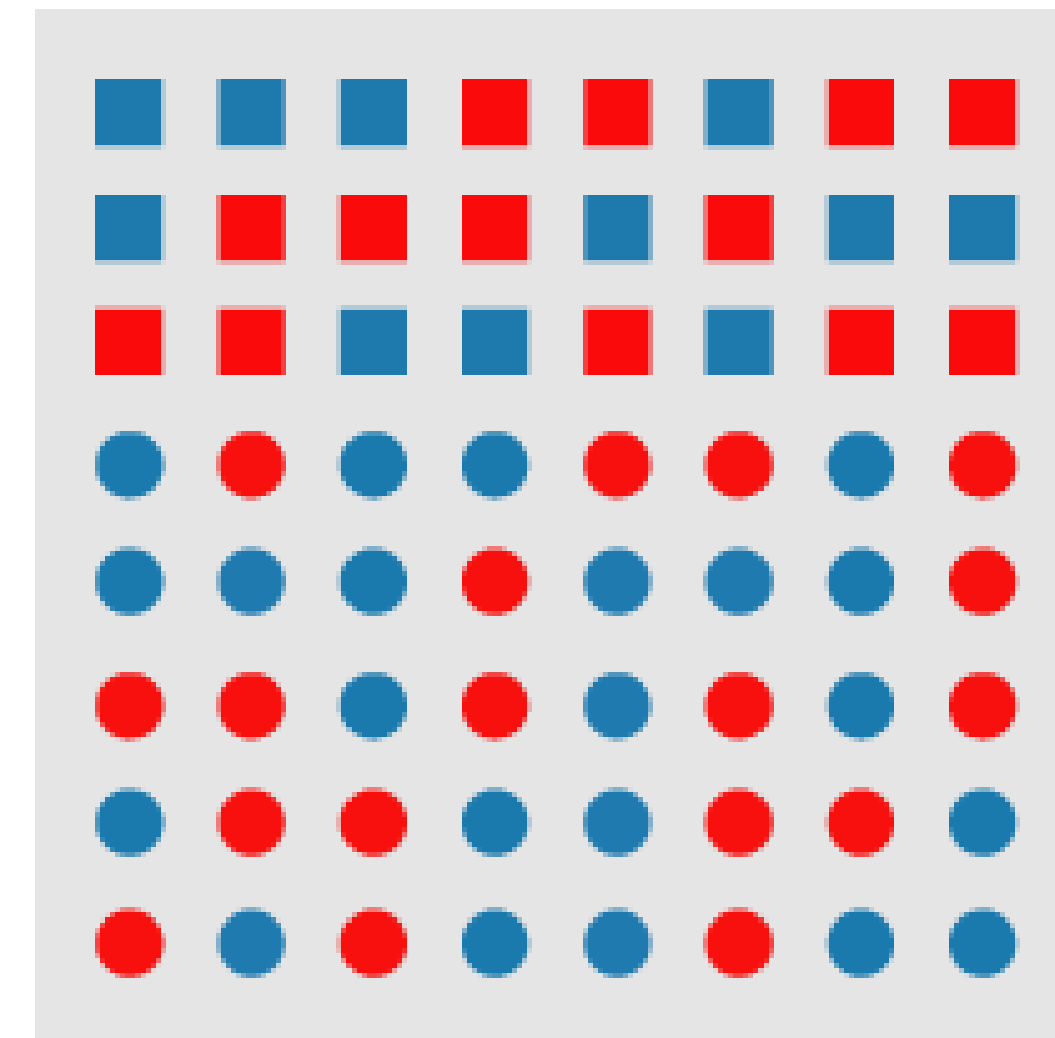
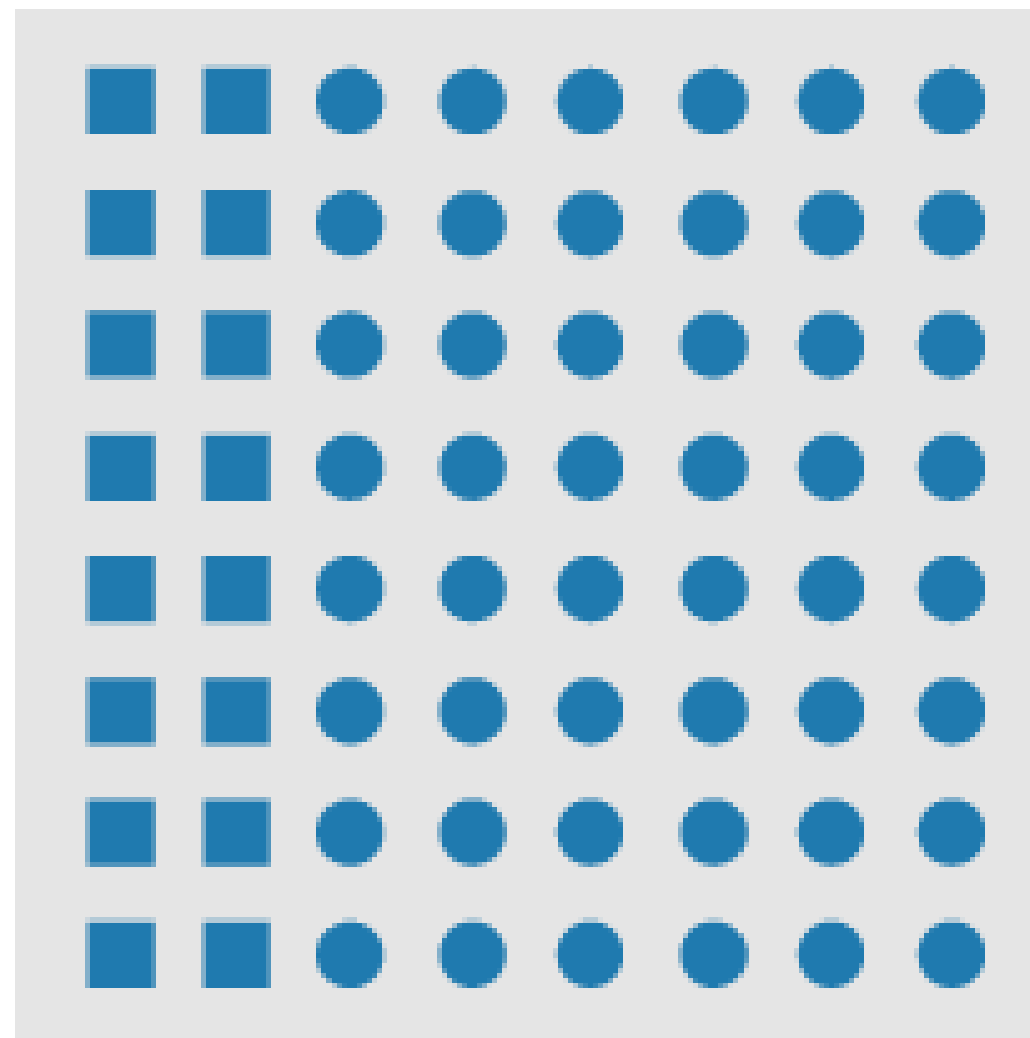
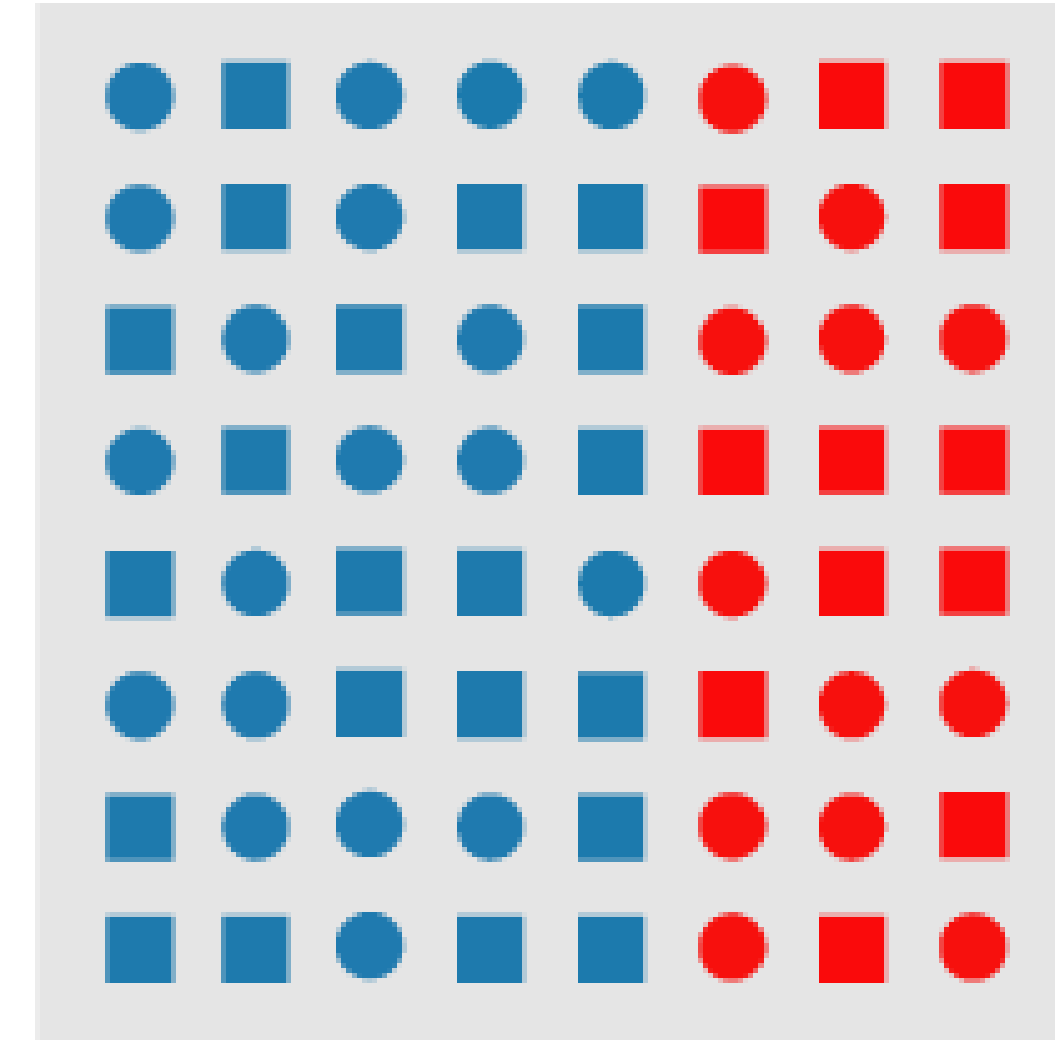
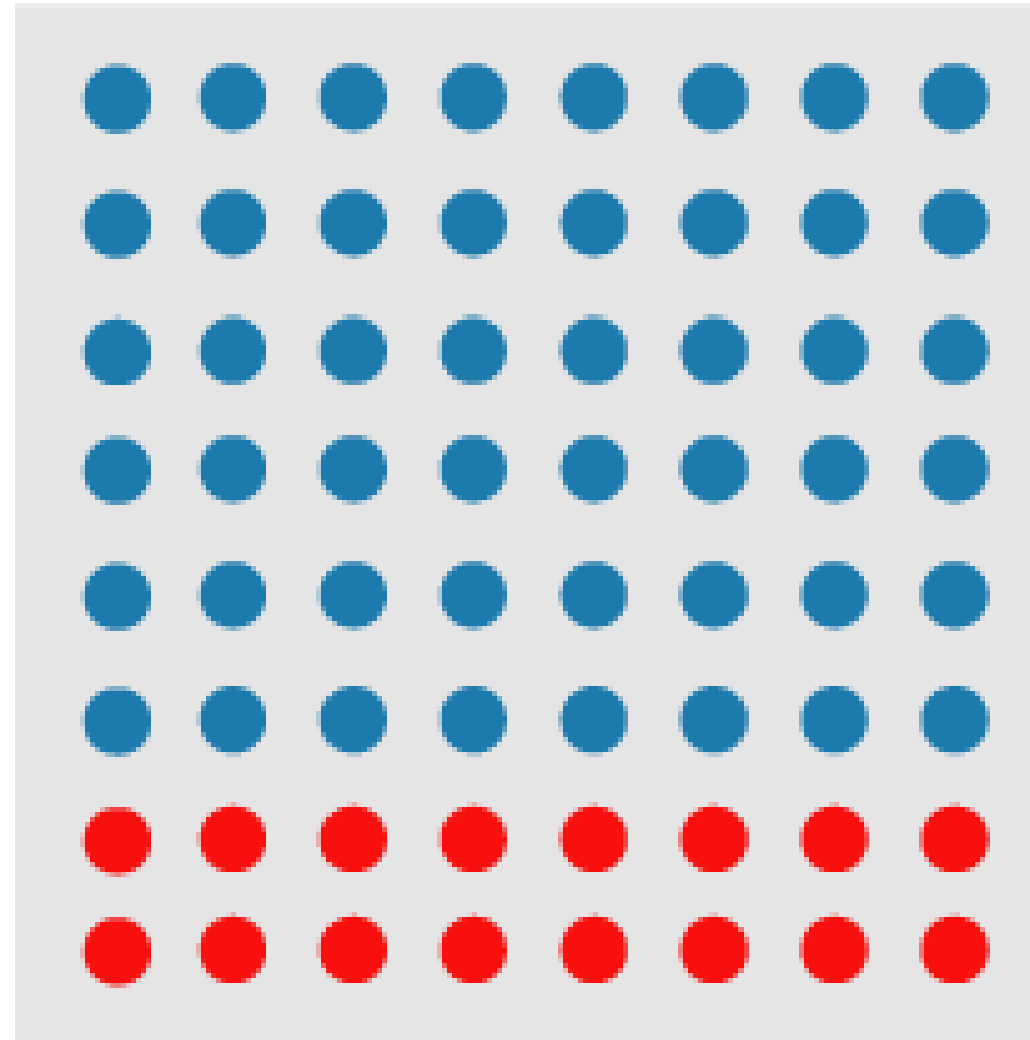
Boundary Detection

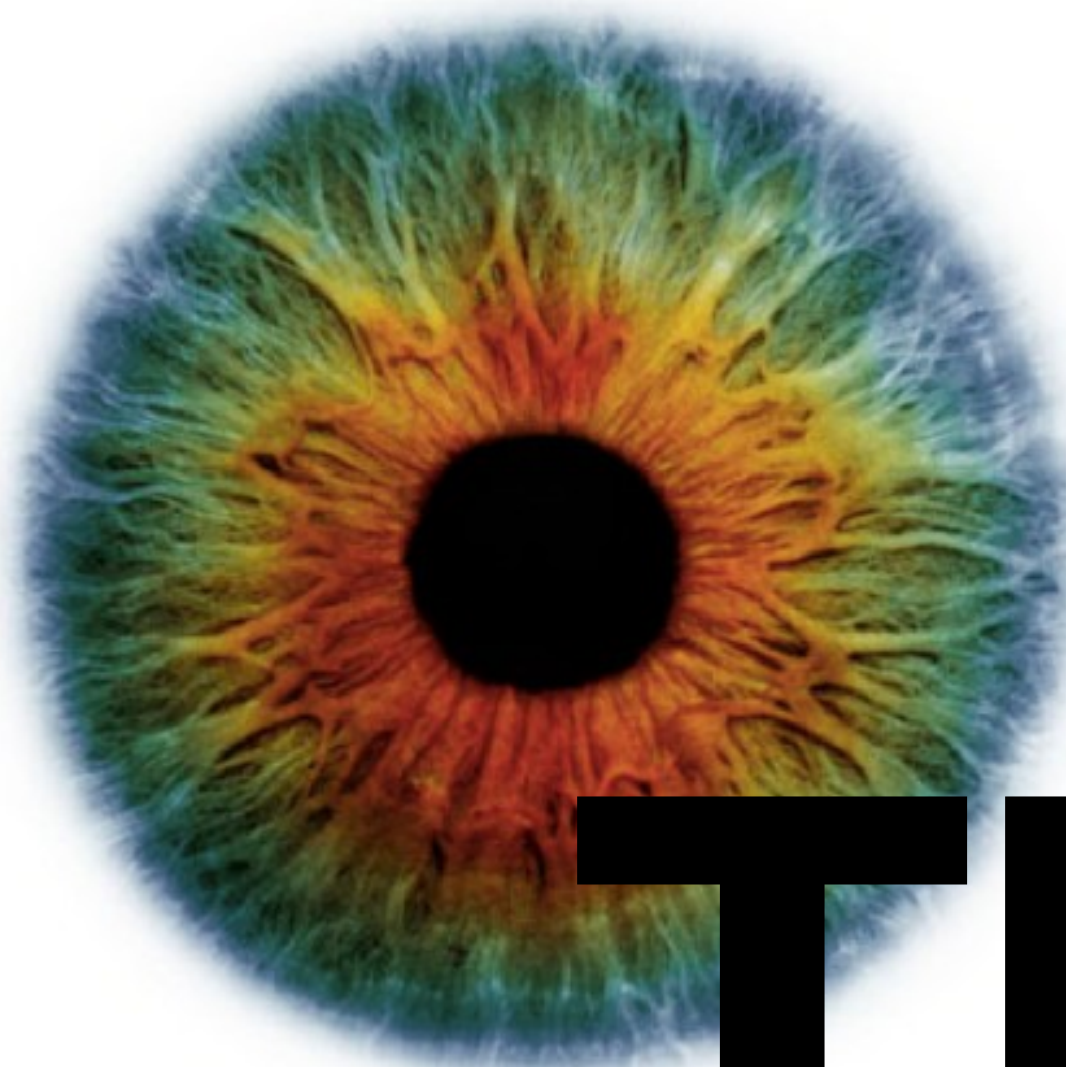


Hierarchy of Pre-attentive Features

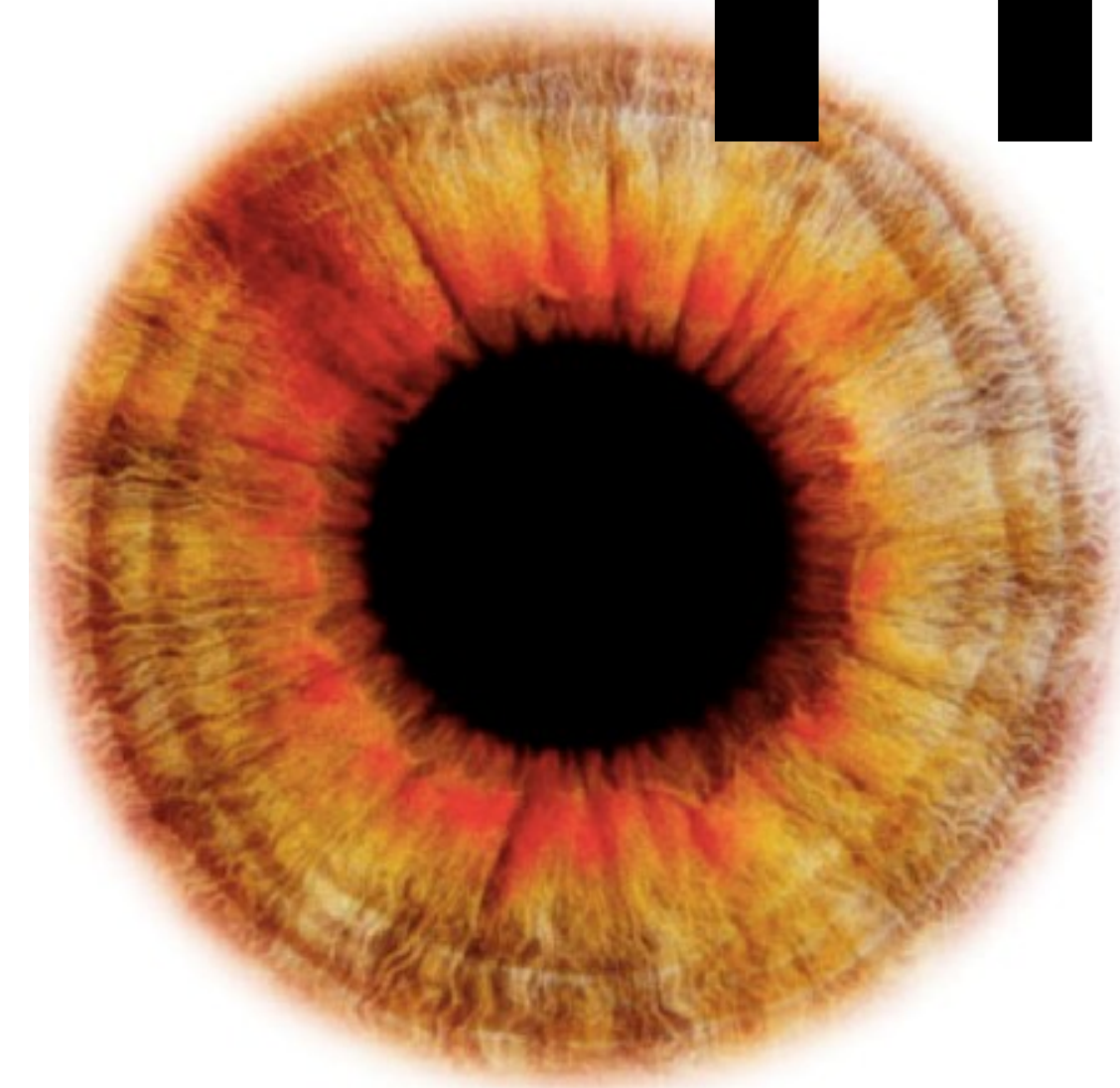


Hierarchy of Pre-attentive Features





The Eye



Human Visual System

Vision works as sequence of fixations and saccades

fixations: maintaining gaze on single location (200-600 ms)

saccades: moving between different
locations (20-100 ms)

Vision not similar to a camera

More similar to a dynamic and
ongoing construction project

Human Visual System

Vision works as sequence of **fixations** and **saccades**

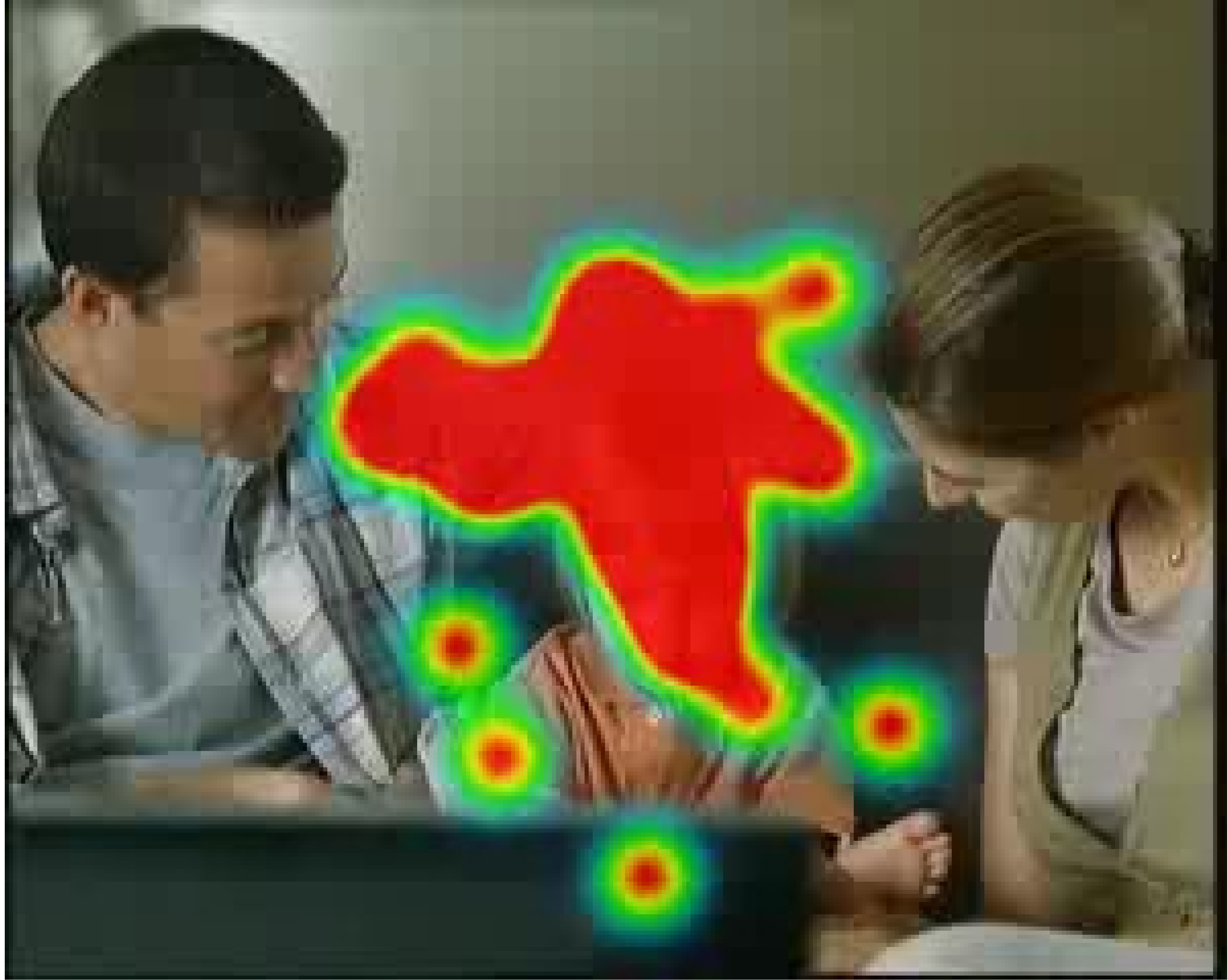
fixations: maintaining gaze on single location (200-600 ms)

saccades: moving between different locations (20-100 ms)

Vision not similar to a camera

More similar to a dynamic and ongoing construction project



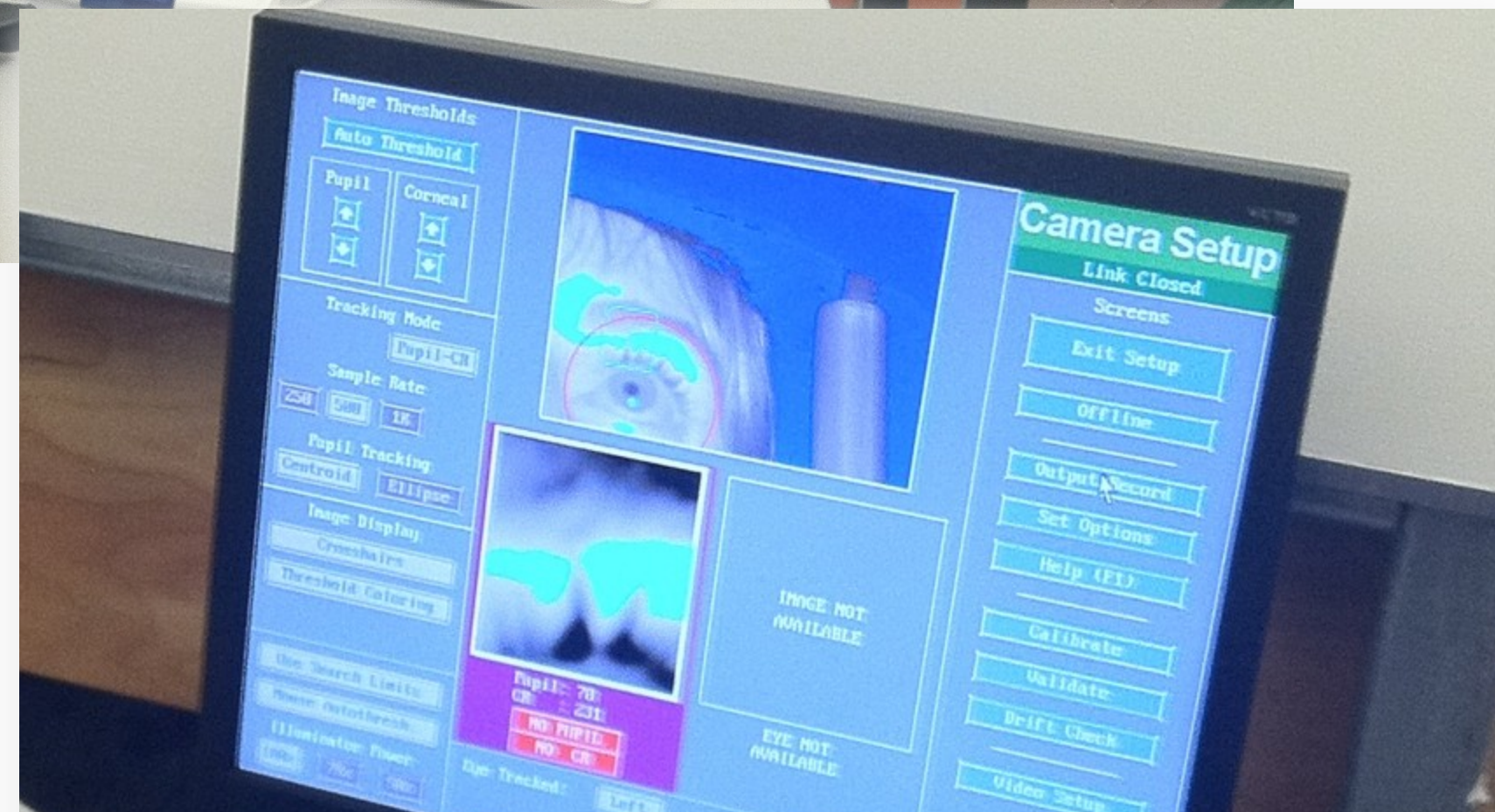


Eye-Tracking Experiment

Visualization display

Infrared camera

Eye-tracking control computer



SensoMotoric Instruments presents

Karaoke

Gaze Tracking and User Observation

Human Visual System

No general purpose vision

What we see depends on our goals and expectations

Relative judgments: strong

Absolute judgments: weak



Human Visual System

No general purpose vision

What we see depends on our goals and expectations

Relative judgments: strong

Absolute judgments: weak



Ames Room

